

Collaborative Inference and Learning between Edge SLMs and Cloud LLMs: A Survey of Algorithms, Execution, and Open Challenges

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As large language models (LLMs) evolve, deploying them solely in the cloud or compressing them for edge devices has become inadequate due to concerns about latency, privacy, cost, and personalization. This survey explores a collaborative paradigm in which cloud-based LLMs and edge-deployed small language models (SLMs) cooperate across both inference and training. We present a unified taxonomy of edge-cloud collaboration strategies. For inference, we categorize approaches into task assignment, task division, and mixture-based collaboration at both task and token granularity, encompassing adaptive scheduling, resource-aware offloading, speculative decoding, and modular routing. For training, we review distributed adaptation techniques, including parameter alignment, pruning, bidirectional distillation, and small-model-guided optimization. We further summarize datasets, benchmarks, and deployment cases, and highlight privacy-preserving methods and vertical applications. This survey provides the first systematic foundation for LLM-SLM collaboration, bridging system and algorithm co-design to enable efficient, scalable, and trustworthy edge-cloud intelligence.

CCS Concepts: • **Computing methodologies** → **Distributed artificial intelligence**; *Neural networks*; • **Computer systems organization** → Embedded and cyber-physical systems; Cloud computing.

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边缘 SLM 和云 LLM 之间的协作推理和学习：算法、执行和开放挑战的调查

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随着大语言模型 (LLM) 的发展，由于对延迟、隐私、成本和个性化的担忧，仅在云中部署它们或为边缘设备压缩它们已经不够了。这项调查探索了一种协作范例，其中基于云的 LLM 和边缘部署的小语言模型 (SLM) 在推理和训练方面进行合作。我们提出了边缘云协作策略的统一分类。为了进行推理，我们在任务和令牌粒度上将方法分类为任务分配、任务划分和基于混合的协作，包括自适应调度、资源感知卸载、推测性解码和模块化路由。对于训练，我们回顾了分布式自适应技术，包括参数对齐、修剪、双向蒸馏和小模型引导优化。我们进一步总结了数据集、基准测试和部署案例，并重点介绍了隐私保护方法和垂直应用。这项调查为 LLM-SLM 协作、桥接系统和算法协同设计提供了第一个系统基础，以实现高效、可扩展和值得信赖的边缘云智能。

CCS 概念： • 计算方法→分布式人工智能；神经网络；嵌入式和网络物理系统；云计算。 • 计算机系统组织 →

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1 Introduction

Large language models (LLMs) [139–142] have demonstrated remarkable proficiency across a broad spectrum of natural language processing tasks. However, their substantial computational and memory demands make on-device deployment on resource-constrained edge devices prohibitive [24, 25, 187]. With the rapid advancement of big data, cloud computing, and edge computing, edge-cloud collaborative intelligence has emerged as a promising paradigm for unlocking data value and delivering ubiquitous AI [35, 55, 163, 255]. This paradigm combines the cloud’s computational power and generalization ability with the edge’s responsiveness and adaptability [5, 176].

Rather than merely compressing large models for edge deployment, we focus on the systematic co-optimization of architecture, algorithms, execution, and privacy across cloud-based LLMs and edge-deployed small language models (SLMs) [232, 233, 235, 236]. By leveraging feature sharing, task partitioning, and knowledge transfer, LLMs and SLMs collaborate to deliver efficient and reliable intelligent services in heterogeneous environments [224–226]. Existing cloud-centric methods exploit centralized LLMs for strong generalization but incur high latency, privacy risks, and poor adaptation to localized contexts [57, 197, 247]. In contrast, edge-centric approaches offer fast response and personalization but are limited by on-device compute and generality [25, 118, 237–239]. Prior surveys have examined cloud-centric, edge-centric, and device-to-device cooperative paradigms, yet each faces key limitations in real-world deployment [162]. The LLM-SLM collaborative paradigm unifies these strengths by enabling hierarchical cooperation between large cloud models and small on-device models [6, 14].

Unlike federated learning [47, 279] or model compression [20], this approach treats LLMs and SLMs as distinct but interactive agents [72, 261], enabling more dynamic and modular collaboration across the edge-cloud continuum. However, realizing this vision poses several challenges across both inference and training. From the inference perspective, architectural heterogeneity between LLMs and SLMs complicates unified scheduling and deployment; varying task granularities, resource budgets, and latency constraints make coordinated model execution difficult [228, 230]; and network instability limits high-frequency interactions between models [21], such as model swapping [22] or cross-attention fusion, which are essential for maintaining semantic consistency and responsiveness. On the training side, heterogeneity in data distributions [23], task formulations, and model architectures hinders effective knowledge transfer, particularly for distillation and adaptation methods that assume structural or objective alignment. Moreover, the prevalence of non-IID edge data [159, 279] and the need for personalized model behavior exacerbate this issue, as local fine-tuning may lead to overfitting, while centralized updates risk diluting critical edge-specific or causally relevant patterns [11, 71]. These intertwined factors collectively underscore the need for more principled designs of collaborative learning and inference frameworks between heterogeneous models.

These limitations motivate a dedicated LLM-SLM collaboration framework that is structurally aware, distributionally robust, and capable of delivering efficient, reliable, and generalizable intelligence under dynamic, resource-constrained conditions.

In 2021, Alibaba DAMO Academy and Zhejiang University conducted the first large-scale survey on cloud-edge collaboration [262], identifying three synergy paradigms. In 2022, Alibaba’s Top Ten Technology Trends highlighted “co-evolution between cloud-based large models and edge-side small models”, emphasizing intelligent, privacy-aware end-device services [242]. Qualcomm echoed this in 2023 with Hybrid AI white papers [243, 245], advocating intelligent workload partitioning between LLMs and SLMs to address latency, efficiency, privacy, and personalization. These developments reflect a shift from centralized LLM pipelines to hybrid edge-cloud architectures, supported by emerging industrial-scale frameworks. Walle [264] offers an end-to-end system for development, deployment, and runtime, enabling cloud-edge collaboration across 300+ tasks with over 10 billion daily invocations, powering personalized recommendations, multi-modal understanding, and real-time 3D rendering via a lightweight edge inference engine. Luoxi [265] adopts a “slow-fast” learning strategy, where cloud LLMs generate latent representations to assist fast, personalized inference on edge devices with real-time

1 简介

大语言模型 (LLM) [139-142] 在广泛的自然语言处理任务中表现出了卓越的熟练程度。然而，它们大量的计算和内存需求使得在资源受限的边缘设备上部署变得令人望而却步[24,25,187]。随着大数据、云计算和边缘计算的快速发展，边云协同智能已成为释放数据价值和提供无处不在的人工智能的有前景的范例[35,55,163,255]。这种范例将云的计算能力和泛化能力与边缘的响应能力和适应性结合起来[5, 176]。

我们不仅仅压缩边缘部署的大型模型，而是专注于跨基于云的 LLM 和边缘部署的小语言模型 (SLM) 的架构、算法、执行和隐私的系统协同优化 [232,233,235,236]。通过利用功能共享、任务划分和知识转移，LLM 和 SLM 协作在异构环境中提供高效、可靠的智能服务 [224-226]。现有的以云为中心的方法利用集中式大语言模型来实现强泛化，但会带来高延迟、隐私风险和对本地环境的适应性较差[57,197,247]。相比之下，以边缘为中心的方法提供快速响应和个性化，但受到设备上计算和通用性的限制[25,118,237-239]。之前的调查已经研究了以云为中心、以边缘为中心和设备到设备的协作范例，但每种范例在实际部署中都面临着关键限制[162]。LLM-SLM 协作范例通过实现大型云模型和小型设备上模型之间的分层协作来统一这些优势 [6, 14]。

与联邦学习 [47, 279] 或模型压缩 [20] 不同，这种方法将 LLM 和 SLM 视为不同但交互式的代理 [72, 261]，从而在边缘云连续体中实现更加动态和模块化的协作。然而，实现这一愿景在推理和训练方面都面临着一些挑战。从推理角度来看，LLM和SLM之间的架构异构使得统一调度和部署变得复杂；不同的任务粒度、资源预算和延迟限制使得协调模型执行变得困难[228, 230]；网络不稳定性限制了模型之间的高频交互[21]，例如模型交换[22]或交叉注意力融合，这对于维持语义一致性和响应能力至关重要。在训练方面，数据分布[23]、任务表述和模型架构的异质性阻碍了有效的知识转移，特别是对于假设结构或目标对齐的蒸馏和适应方法。此外，非独立同分布边缘数据的盛行 [159, 279] 以及个性化模型行为的需求加剧了这个问题，因为局部微调可能导致过度拟合，而集中更新则有可能稀释关键的边缘特定或因果相关模式 [11, 71]。这些相互交织的因素共同强调需要对异构模型之间的协作学习和推理框架进行更有原则的设计。

这些限制催生了专用的 LLM-SLM 协作框架，该框架具有结构感知、分布式稳健性，并且能够在动态、资源受限的条件下提供高效、可靠和通用的情报。

2021年，阿里巴巴达摩院和浙江大学进行了首次大规模的云边协同调查[262]，确定了三种协同范式。2022年，阿里巴巴十大技术趋势强调“云端大模型与边缘小模型的共同进化”，强调智能、隐私感知的终端服务[242]。高通在 2023 年发布了混合 AI 白皮书 [243, 245] 呼应了这一点，提倡在 LLM 和 SLM 之间进行智能工作负载划分，以解决延迟、效率、隐私和个性化问题。这些发展反映了从集中式大语言模型管道向混合边缘云架构的转变，并得到新兴工业规模框架的支持。Walle [264] 提供了一个用于开发、部署和运行时的端到端系统，支持跨 300+ 任务的云边缘协作，每日调用次数超过 100 亿次，通过轻量级边缘推理引擎支持个性化推荐、多模式理解和实时 3D 渲染。Luoxi [265]采用“慢-快”学习策略，其中云大语言模型生成潜在表示，以协助在边缘设备上实时进行快速、个性化的推理

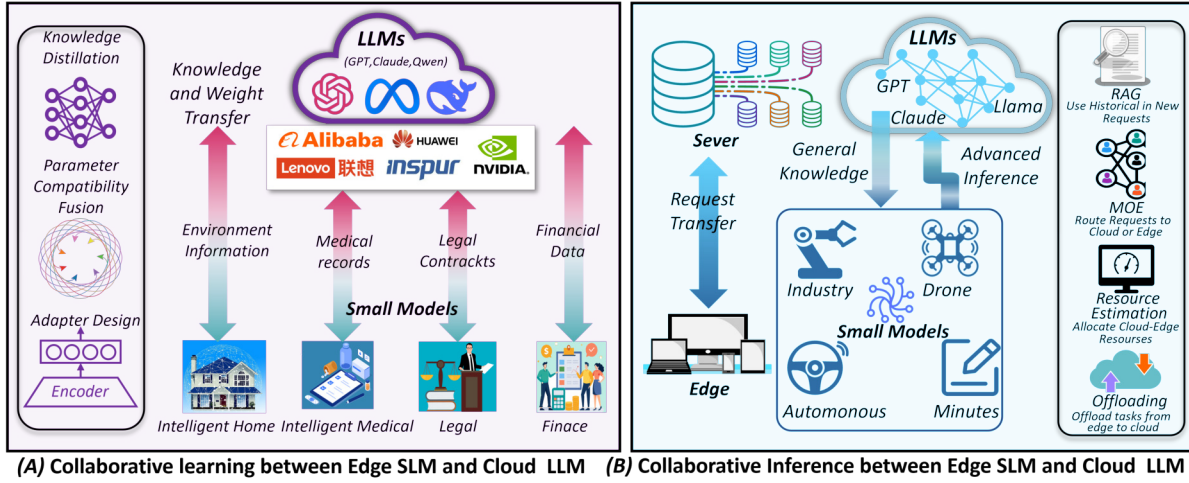


Fig. 1. Overview of cloud-edge collaboration workflows: (a) training and (b) inference.

feedback. InfiGUIAgent [266] demonstrates hierarchical edge reasoning through two-stage fine-tuning for GUI interaction in multi-modal scenarios. Real-world applications further underscore this paradigm's necessity: in vision and graphics (urban video analytics [184], autonomous driving [55, 56], XR), in language tasks (virtual assistants [240, 241], input methods, dialogue agents), and in personalized recommendation [30, 63, 221]. Edge intelligence is also expanding into multi-modal and sensor-based domains such as medical diagnosis [225], smart homes [204], and industrial monitoring, where energy efficiency and responsiveness are critical. Together, these academic and industrial efforts highlight both the technical feasibility and practical urgency of cloud-edge LLM-SLM collaboration, motivating a comprehensive survey of existing paradigms, system designs, and open challenges.

1.1 Related Surveys and Their Scope

Xu *et al.* [25] provide the first systematic overview of the full AIGC service lifecycle, from data collection and training to inference, and propose a collaborative cloud-edge-end infrastructure for mobile networks. Qu *et al.* [4] review LLM deployment at the mobile network edge, emphasizing resource-efficient techniques, architectural design, and edge LLM caching. In contrast, Xu *et al.* [246] prioritize system-level optimization, such as model compactification and token pruning, to accelerate inference and enhance deployment efficiency. For task offloading and resource allocation, Wang *et al.* [250] model task migration in mobile-cloud offloading, while [249] apply reinforcement learning to scheduling in CETCN scenarios. Xu *et al.* [248] survey decentralized continual learning on distributed devices, categorizing three algorithmic strategies for mitigating catastrophic forgetting and distributional shift. Building on this, Niu *et al.* [163] comprehensively review collaborative learning mechanisms between device-side small models and cloud-based large models from system, algorithmic, and application perspectives. Under constrained communication, Zhang *et al.* [24] propose air-ground collaborative inference strategies, and Lin *et al.* [187] introduce a multi-agent LLM framework for natural language tasks in 6G networks. Chen *et al.* [162] systematically summarize LLM-SLM collaboration paradigms, pipelining, routing, distillation, and fusion, as foundational strategies for efficient, adaptive intelligent systems. Existing surveys tend to focus narrowly on system deployment or isolated techniques (*e.g.*, distillation, pipelining) and lack a unified algorithmic

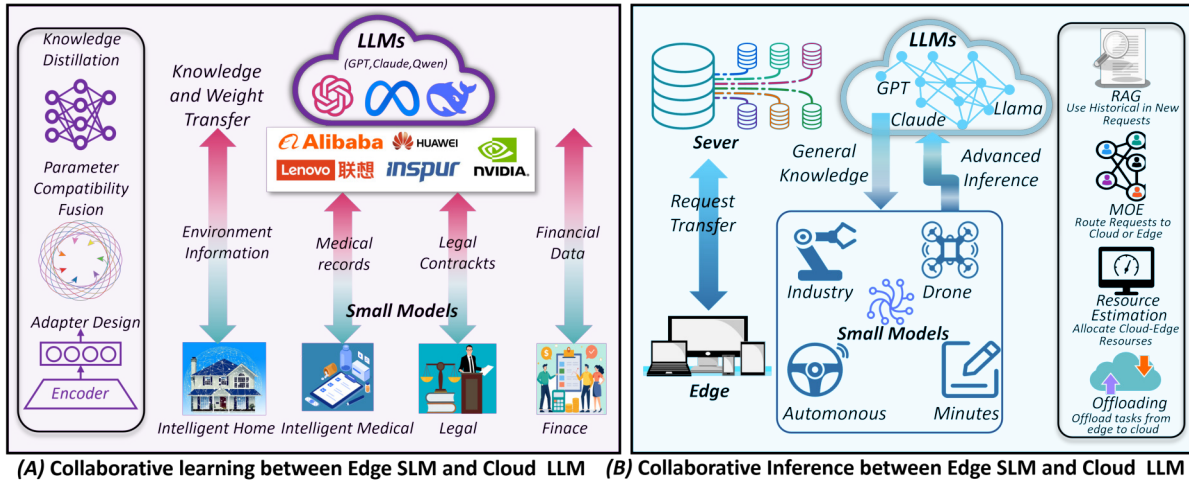


图 1. 云边协作工作流程概述：(a) 训练和 (b) 推理。

反馈。InfiGUIAgent [266]通过两阶段微调演示了多模式场景中 GUI 交互的分层边缘推理。现实世界的应用进一步强调了这种范式的必要性：在视觉和图形领域（城市视频分析[184]、自动驾驶[55, 56]、XR）、语言任务（虚拟助手[240, 241]、输入方法、对话代理）以及个性化推荐[30, 63, 221]。边缘智能还扩展到多模式和基于传感器的领域，例如医疗诊断[225]、智能家居[204]和工业监控，其中能源效率和响应能力至关重要。这些学术和工业界的努力共同凸显了云边缘 LLM-SLM 协作的技术可行性和实际紧迫性，激发了对现有范式、系统设计和开放挑战的全面调查。

1.1 相关调查及其范围

徐等人。[25]首次系统概述了整个 AIGC 服务生命周期，从数据收集和训练到推理，并提出了用于移动网络的协作云边基础设施。曲等人。[4]回顾移动网络边缘的 LLM 部署，强调资源高效技术、架构设计和边缘 LLM 缓存。相比之下，徐等人。[246]优先考虑系统级优化，例如模型压缩和令牌修剪，以加速推理并提高部署效率。对于任务卸载和资源分配，Wang 等人。[250]对移动云卸载中的任务迁移进行建模，而[249]将强化学习应用于 CETCN 场景中的调度。徐等人。[248]调查了分布式设备上的去中心化持续学习，对减轻灾难性遗忘和分布转移的三种算法策略进行了分类。在此基础上，Niu 等人。[163]从系统、算法和应用的角度全面回顾了设备侧小模型和基于云的大模型之间的协作学习机制。在沟通受限的情况下，Zhang 等人。[24]提出了空地协同推理策略，Lin 等人。[187]引入了用于 6G 网络中自然语言任务的多代理 LLM 框架。陈等人。[162]系统地总结了 LLM-SLM 协作范例、流水线、路由、蒸馏和融合，作为高效、自适应智能系统的基础策略。现有的调查往往狭隘地关注系统部署或孤立的技术（例如蒸馏、管道），并且缺乏统一的算法

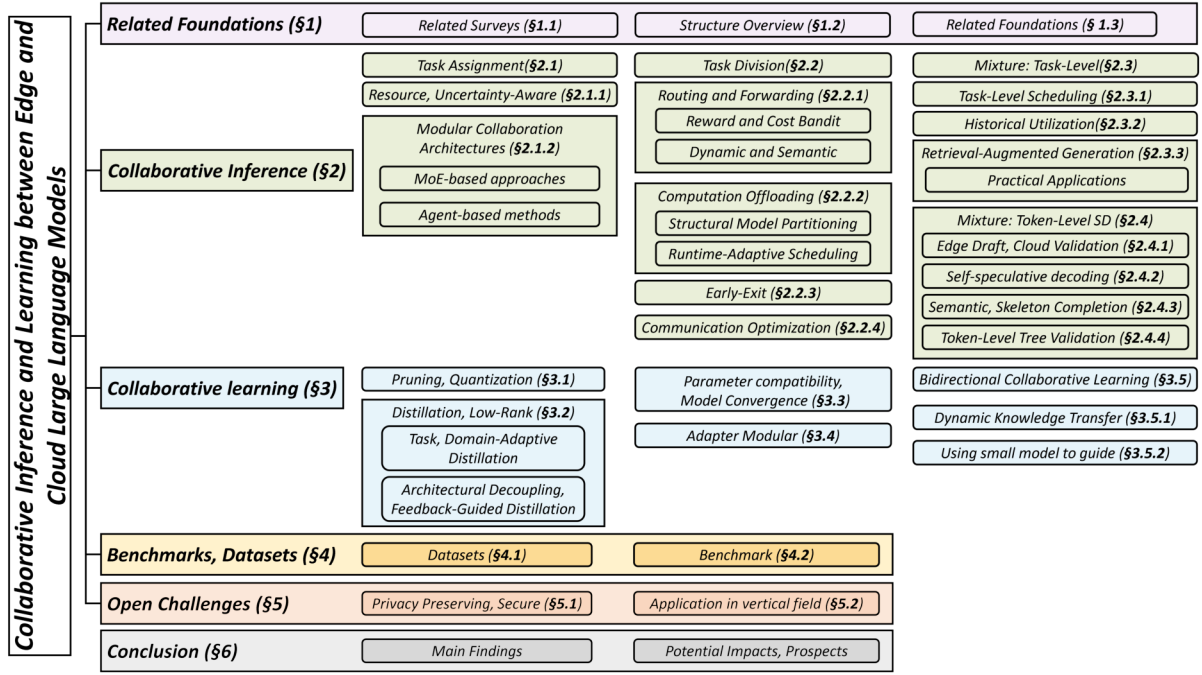


Fig. 2. Overview of collaborative paradigms between edge-cloud large and SLMs, structured along two axes: inference cooperation and training collaboration.

perspective on collaborative large-small model training and inference. Crucially, none abstract collaboration into a structured design space or offer a taxonomy capturing functional roles and interaction boundaries between heterogeneous models in edge-cloud settings.

In contrast, this survey presents the first comprehensive review that jointly considers inference-time collaboration and training-time coordination between LLMs and SLMs. As shown in Fig. 1 and Table 1, we propose a unified taxonomy of collaboration paradigms, encompassing task assignment, task division, and mixture (with task-level and token-level granularities), and analyze their alignment with algorithmic principles and system constraints. On the training side, we summarize collaborative adaptation methods, including bidirectional distillation, quantization, pruning, and low-rank approximation, which facilitate efficient SLM deployment without sacrificing performance. This survey places particular emphasis on the interaction between LLMs and SLMs across edge and cloud environments. By jointly analyzing collaboration across both inference and training phases, our review bridges algorithmic design and deployment needs, offering methodological insights and practical implications for future edge-cloud LLM systems. Moreover, we ground our discussion in recent literature published since 2023, ensuring both coverage and relevance. By jointly addressing both inference and training dimensions, our survey fills a key methodological gap and establishes a generalizable framework for designing large-small model cooperation in heterogeneous, resource-constrained environments.

1.2 Research Questions and Survey Structure

This survey systematically reviews the development, challenges, and future directions of edge-cloud collaborative inference and training with large and SLMs. We organize the discussion around the following core research questions (RQs):

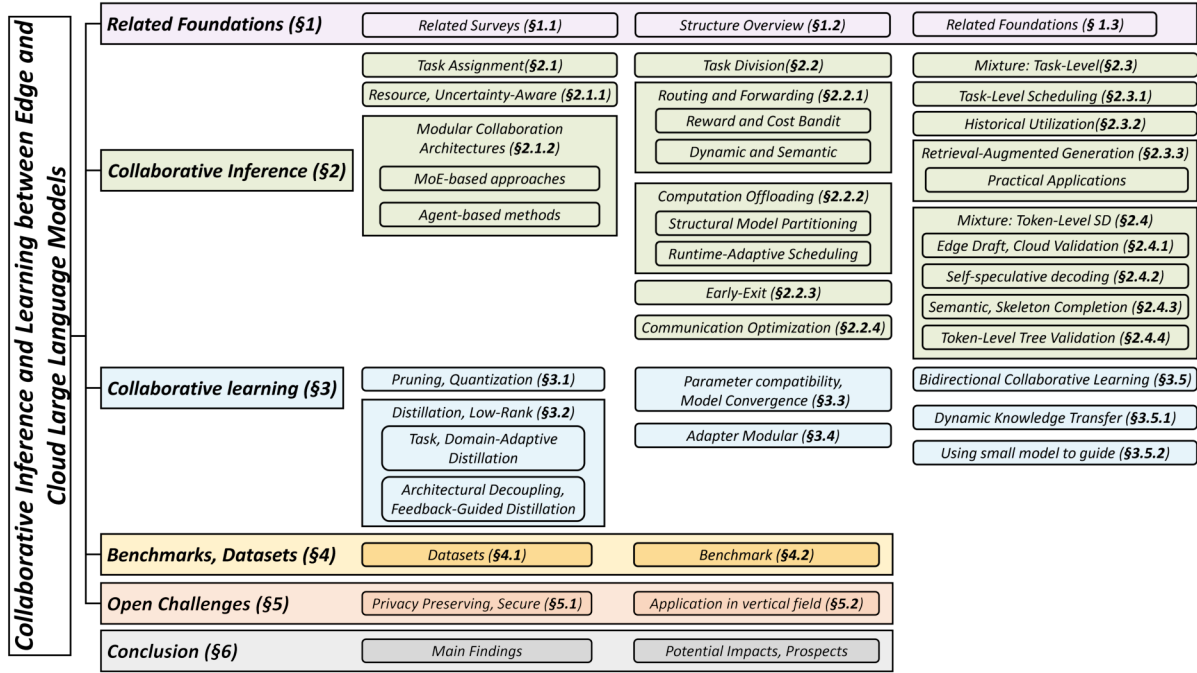


图 2. 大型边缘云和 SLM 之间的协作范例概述，沿两个轴构建：推理协作和训练协作。

对协作大小模型训练和推理的看法。至关重要的是，没有一个将协作抽象到结构化设计空间中，也没有提供捕获边缘云设置中异构模型之间的功能角色和交互边界的分类法。

相比之下，这项调查首次全面审查了大语言模型和 SLM 之间的推理时间协作和训练时间协调。如图1和表1所示，我们提出了协作范式的统一分类，包括任务分配、任务划分和混合（具有任务级和令牌级粒度），并分析它们与算法原理和系统约束的一致性。在训练方面，我们总结了协作适应方法，包括双向蒸馏、量化、剪枝和低秩近似，这些方法有助于在不牺牲性能的情况下高效部署 SLM。这项调查特别强调大语言模型和 SLM 之间跨边缘和云环境的交互。通过共同分析推理和训练阶段的协作，我们的审查将算法设计和部署需求联系起来，为未来的边缘云 LLM 系统提供方法论见解和实际影响。此外，我们的讨论以 2023 年以来发表的最新文献为基础，确保覆盖范围和相关性。通过共同解决推理和训练维度，我们的调查填补了一个关键的方法论空白，并建立了一个在异构、资源受限的环境中设计大小模型合作的通用框架。

1.2 研究问题和调查结构

本调查系统回顾了大型 SLM 的边云协作推理和训练的发展、挑战和未来方向。我们围绕以下核心研究问题 (RQ) 组织讨论：

Table 1. Comparison of this survey with related reviews

Reference	Scope	Focus Area	Distinguishing Features
Xu <i>et al.</i> [25]	AIGC service lifecycle	Mobile network infrastructure design	Proposes end-cloud-edge architecture; omits model-level inference collaboration.
Xu <i>et al.</i> [246]	LLM system deployment	Inference optimization (compression, pruning)	Emphasizes hardware/system acceleration; ignores LLM-SLM interaction logic.
Wang <i>et al.</i> [249, 250]	Mobile-cloud task offloading	Scheduling and resource allocation	Models offloading strategies via RL; does not address collaborative inference.
Xu <i>et al.</i> [248]	Decentralized continual learning	Distributed algorithmic adaptation	Surveys lifelong learning; lacks explicit LLM-SLM collaboration structure.
Niu <i>et al.</i> [163]	LLM-SLM collaborative training	System, algorithm, application layers	Reviews collaborative training; does not cover inference across modalities/tasks.
Zhang <i>et al.</i> [24, 187]	LLMs in constrained networks (6G)	Multi-agent and aerial-ground collaboration	Focuses on special network conditions; proposes communication-specific inference strategies.
Chen <i>et al.</i> [162]	LLM-SLM collaboration mechanisms	Pipelining, routing, distillation, fusion	Summarizes interaction methods; lacks a unified algorithmic design space for inference.
This Survey	Heterogeneous LLM-SLM collaboration	Inference and training algorithms	First to cover end-cloud collaborative inference (task assignment/division/mixture) and training (parameter fusion, knowledge transfer).

- **RQ1** *What is the edge-cloud collaborative inference paradigm with large and small models? What are its fundamental concepts and system architecture?* We focus on the basic paradigm, collaboration roles, and system-level design.
- **RQ2** *What are the major paradigms and collaboration patterns in edge-cloud inference?* We cover task scheduling, mixture cooperation at task and token levels, and speculative decoding. Examines how collaboration reduces latency, adapts to uncertainty, and improves efficiency.
- **RQ3** *Why and how should we study collaborative training between large and small models in edge-cloud environments?* We explore the necessity of training collaboration under heterogeneity, summarizing paradigms such as distillation, modular tuning, and parameter alignment for cross-device adaptation.
- **RQ4** *Why is it important to review benchmarks, privacy-preserving methods, and vertical applications in the context of edge-cloud collaboration?* We address fair evaluation standards, privacy challenges in training and inference, and practical requirements across real-world domains.

As shown in Fig. 2, we structure the survey as follows:

表 1. 本次调查与相关评论的比较

Reference	Scope	Focus Area	Distinguishing Features
Xu <i>et al.</i> [25]	AIGC service lifecycle	Mobile network infrastructure design	Proposes end-cloud-edge architecture; omits model-level inference collaboration.
Xu <i>et al.</i> [246]	LLM system deployment	Inference optimization (compression, pruning)	Emphasizes hardware/system acceleration; ignores LLM-SLM interaction logic.
Wang <i>et al.</i> [249, 250]	Mobile-cloud task offloading	Scheduling and resource allocation	Models offloading strategies via RL; does not address collaborative inference.
Xu <i>et al.</i> [248]	Decentralized continual learning	Distributed algorithmic adaptation	Surveys lifelong learning; lacks explicit LLM-SLM collaboration structure.
Niu <i>et al.</i> [163]	LLM-SLM collaborative training	System, algorithm, application layers	Reviews collaborative training; does not cover inference across modalities/tasks.
Zhang <i>et al.</i> [24, 187]	LLMs in constrained networks (6G)	Multi-agent and aerial-ground collaboration	Focuses on special network conditions; proposes communication-specific inference strategies.
Chen <i>et al.</i> [162]	LLM-SLM collaboration mechanisms	Pipelining, routing, distillation, fusion	Summarizes interaction methods; lacks a unified algorithmic design space for inference.
This Survey	Heterogeneous LLM-SLM collaboration	Inference and training algorithms	First to cover end-cloud collaborative inference (task assignment/division/mixture) and training (parameter fusion, knowledge transfer).

- RQ1 什么是大模型、小模型的边云协同推理范式？它的基本概念和系统架构是什么？我们专注于基本范式、协作角色和系统级设计。
- RQ2 边云推理的主要范式和协作模式是什么？我们涵盖任务调度、任务和令牌级别的混合合作以及推测解码。检查协作如何减少延迟、适应不确定性并提高效率。
- RQ3 为什么以及如何研究边云环境中大小模型之间的协同训练？我们探讨了异构下训练协作的必要性，总结了蒸馏、模块化调整和跨设备适应的参数对齐等范式。
- RQ4 为什么在边缘云协作的背景下审查基准、隐私保护方法和垂直应用程序很重要？我们解决公平的评估标准、训练和推理中的隐私挑战以及现实世界领域的实际要求。

如图 2 所示，我们的调查结构如下：

- **Section 1.3** introduces the fundamentals of edge-cloud collaboration with large and SLMs, covering system architectures, edge device constraints, and trends in large-model development (addresses RQ1).
- **Section 2** examines task assignment and inference strategies, with emphasis on dynamic scheduling and draft-refine verification frameworks, illustrating how edge and cloud models execute inference collaboratively and efficiently (addresses RQ2).
- **Section 3** surveys cloud-edge training architectures and deployment strategies, including quantization, pruning, parameter-efficient fine-tuning, and bidirectional knowledge transfer, to support efficient SLM deployment without sacrificing performance (addresses RQ3).
- **Section 4 - Section 5** summarize benchmarks, evaluation metrics, and domain-specific deployments to facilitate model comparison and practical implementation, and highlight open challenges such as generalizability, privacy, sustainability, and multi-agent collaboration (addresses RQ4).
- **Section 6** concludes with key insights and future research directions.

1.3 Related Concepts and Foundations

1.3.1 Definition and Characteristics of LLMs. LLMs typically refer to transformer-based architectures with billions to trillions of parameters. These models are pretrained on massive-scale corpora using strategies such as autoregressive learning (e.g., GPT [141, 240]) or masked language modeling (e.g., Qwen [235], DeepSeek [289]). Architecturally, LLMs stack tens to hundreds of transformer blocks, each comprising multi-head attention, feed-forward layers, residual connections, and normalization. As scale grows, LLMs exhibit emergent abilities, such as symbolic reasoning, instruction following, and multimodal understanding, that do not scale linearly with parameter count. In edge-cloud collaboration, LLMs are typically cloud-hosted due to their intensive computational and memory needs. While offering strong generalization and zero-/few-shot transfer across tasks, they also introduce significant communication overhead, as each inference often requires multiple cloud interactions, increasing latency and bandwidth load. Additionally, their training and fine-tuning involve high I/O costs and poor device adaptability, limiting their use in real-time or privacy-sensitive edge scenarios.

1.3.2 Definition and Characteristics of SLMs. SLMs, typically comprising millions to a few hundred million parameters, are built for resource-constrained environments while retaining core language modeling abilities. They simplify architecture via shallower transformer stacks, smaller hidden sizes, and fewer attention heads, greatly reducing memory and power use, enabling efficient deployment on mobile devices, microcontrollers, and edge nodes. SLMs emphasize responsiveness and local availability over broad reasoning. Though they lack LLMs' emergent abilities, they perform well in context-specific tasks like smart input, offline assistants, and embedded systems. Through knowledge distillation, LoRA fine-tuning, and instruction tuning, SLMs can approximate LLM outputs in narrow domains while maintaining low latency and data locality. In collaborative setups, they act as first-response agents or draft generators, offloading complex reasoning to cloud LLMs only when needed, thus optimizing communication cost and model efficiency.

1.3.3 Complementarity of hardware architectures and application scenarios. The foundation for cloud-edge LLM-SLM collaboration stems from hardware-driven capability partitioning: Cloud servers leverage multi-GPU clusters (e.g., NVIDIA H100) with high-bandwidth interconnects (NVLink/RoCE) to run trillion-parameter models (e.g., LLaMA-3 405B) for complex reasoning and global data processing [193], while resource-constrained edge devices, spanning smartphones (6-16GB RAM), XR headsets, and embedded systems (e.g., Jetson Orin), deploy billion-parameter SLMs (e.g., TinyBERT [157]) for latency-critical tasks (<10ms response) and local data optimization [192]. This architectural complementarity enables hierarchical task allocation: SLMs execute real-time perception (keyword detection, speech-to-text), privacy-sensitive preprocessing (medical data anonymization), and intent filtering (90% accuracy in customer service), offloading complex generation/reasoning to LLMs.

- 1.3 节介绍了边缘云与大型模型和 SLM 协作的基础知识，涵盖系统架构、边缘设备约束以及大型模型开发的趋势（解决 RQ1）。
- 第 2 节研究任务分配和推理策略，重点是动态调度和草稿细化验证框架，说明边缘和云模型如何协作且高效地执行推理（解决 RQ2）。
- 第 3 部分调查了云边缘训练架构和部署策略，包括量化、修剪、参数高效微调和双向知识传输，以支持高效的 SLM 部署而不牺牲性能（解决 RQ3）。
- 第 4 节 - 第 5 节总结了基准、评估指标和特定领域的部署，以促进模型比较和实际实施，并强调普遍性、隐私、可持续性和多代理协作等开放挑战（解决 RQ4）。
- 第六部分总结了主要见解和未来的研究方向。

1.3 相关概念和基础

1.3.1 大语言模型的定义和特点。大语言模型通常指的是具有数十亿到数万亿参数的基于变压器的架构。这些模型使用自回归学习（例如 GPT [141, 240]）或屏蔽语言建模（例如 Qwen [235]、DeepSeek [289]）等策略在大规模语料库上进行预训练。在架构上，LLM 堆叠了数十到数百个转换器块，每个块都包含多头注意力、前馈层、残差连接和归一化。随着规模的扩大，大语言模型表现出新兴的能力，例如符号推理、指令遵循和多模式理解，这些能力不随参数数量线性扩展。在边缘云协作中，由于计算和内存需求密集，大语言模型通常由云托管。在提供强大的泛化能力和跨任务的零/少镜头传输的同时，它们也带来了巨大的通信开销，因为每个推理通常需要多次云交互，从而增加了延迟和带宽负载。此外，它们的训练和微调涉及较高的 I/O 成本和较差的设备适应性，限制了它们在实时或隐私敏感的边缘场景中的使用。

1.3.2 SLM 的定义和特征。SLM 通常包含数百万到数亿个参数，专为资源受限的环境而构建，同时保留核心语言建模能力。它们通过更浅的变压器堆栈、更小的隐藏尺寸和更少的注意力头来简化架构，从而大大减少内存和功耗，从而实现在移动设备、微控制器和边缘节点上的高效部署。SLM 强调响应能力和本地可用性，而不是广泛的推理。尽管他们缺乏大语言模型的应急能力，但他们在智能输入、离线助手和嵌入式系统等特定环境任务中表现良好。通过知识蒸馏、LoRA 微调和指令调整，SLM 可以在窄域中逼近 LLM 输出，同时保持低延迟和数据局部性。在协作设置中，它们充当第一响应代理或草稿生成器，仅在必要时将复杂的推理卸载给云大语言模型，从而优化通信成本和模型效率。

1.3.3 硬件架构和应用场景的互补。云边缘 LLM-SLM 协作的基础源于硬件驱动的能力划分：云服务器利用多 GPU 集群（例如 NVIDIA H100）和高带宽互连（NVLink/RoCE）来运行万亿参数模型（例如 LLaMA-3 405B）以进行复杂推理和全局数据处理 [193]，而资源受限的边缘设备则涵盖智能手机（6-16GB RAM）、XR 耳机和嵌入式系统（例如 Jetson Orin），部署十亿参数的 SLM（例如 TinyBERT [157]）来执行延迟关键型任务（<10ms 响应）和本地数据优化 [192]。这种架构上的互补性实现了分层任务分配：SLM 执行实时感知（关键词检测、语音转文本）、隐私敏感预处理（医疗数据匿名化）和意图过滤（客户服务准确率达 90%），将复杂的生成/推理工作卸载给 LLM。

Table 2. Comparison of major paradigms in cloud-edge collaborative inference

Category	Definition & Characteristics	Advantages	Limitations
Task assignment § 2.1	Routes requests to SLM or LLM based on confidence, resource, or task type.	Simple and fast; minimal communication overhead.	Hard switching may misroute uncertain inputs; lacks joint reasoning.
Task division § 2.2	Splits tasks into modules (e.g., early exit, routing) processed by SLM and LLM.	Supports modular, adaptive inference with fine-grained control.	Requires explicit segmentation; incurs coordination overhead.
Mixture: Task-level § 2.3	Combines assignment and division across stages or roles via communication.	Flexible decomposition; leverages complementary strengths.	Dependent on partitioning quality and coordination efficacy.
Mixture: Token-level § 2.4	Collaborates at token generation (e.g., speculative decoding).	Low-latency with accurate output; efficient cloud fallback.	Sensitive to draft quality; complex fusion and validation.

Technical synergies like knowledge distillation (TinyBERT achieving 96% of BERT’s performance [157]) and parameter-efficient tuning (LoRA boosting medical QA accuracy by 20%) further bridge capability gaps, with federated learning ensuring privacy when SLMs transmit anonymized features to cloud LLMs for multimodal fusion, collectively enabling 40% efficiency gains in applications like in-vehicle voice systems (200ms latency) and healthcare diagnostics.

2 Overview of Collaborative Inference

Collaborative inference between edge-deployed SLMs and cloud-based LLMs seeks to optimize latency and service efficiency under bandwidth and responsiveness constraints [288]. As shown in Table 2, recent work has explored three complementary collaboration paradigms. It is important to note that the classification presented here focuses specifically on inference-time collaboration between heterogeneous models. Although some mechanisms, such as mixture-of-experts (MoE), are applicable to both inference and training phases, this survey confines its scope to inference-oriented scenarios. That is, we summarize and compare collaborative strategies that are either designed for or primarily evaluated under inference workloads. We categorize and review existing works from the following perspectives:

- **Task assignment**, which routes queries to the most suitable model (e.g., via agent-based or MoE strategies);
- **Task division**, which decomposes execution across models or system components (e.g., offloading, request routing, or early-exit mechanisms);
- **Mixture strategies**, which integrate assignment and division at task-level or token-level granularity using techniques such as speculative decoding and multi-stage validation.

2.1 Task Assignment: Dynamic Model Selection for Cost-Quality Trade-offs

Task assignment determines whether a request is processed entirely by an edge-side SLM or a cloud-based LLM, in contrast to collaborative decoding, which involves inter-model interaction. To minimize latency and energy consumption under quality-of-service (QoS) constraints, prior methods employ lightweight scorers, calibrated reward estimators, or bandit-based controllers to select the optimal execution path before generation. Some approaches further incorporate user preference vectors or learned model-capability representations for

表2 云边协同推理主要范式比较

Category	Definition & Characteristics	Advantages	Limitations
Task assignment § 2.1	Routes requests to SLM or LLM based on confidence, resource, or task type.	Simple and fast; minimal communication overhead.	Hard switching may misroute uncertain inputs; lacks joint reasoning.
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Mixture: Token-level § 2.4	Collaborates at token generation (e.g., speculative decoding).	Low-latency with accurate output; efficient cloud fallback.	Sensitive to draft quality; complex fusion and validation.

知识蒸馏（TinyBERT 实现了 BERT 性能的 96% [157]）和参数高效调整（LoRA 将医疗 QA 准确性提高了 20%）等技术协同作用进一步缩小了能力差距，联合学习确保了 SLM 将匿名特征传输到云端 LLM 进行多模态融合时的隐私，从而使车载语音系统（200ms 延迟）和医疗诊断等应用的效率提高了 40%。

2 协同推理概述

边缘部署的 SLM 和基于云的 LLM 之间的协作推理旨在在带宽和响应能力限制下优化延迟和服务效率 [288]。如表 2 所示，最近的工作探索了三种互补的协作范例。值得注意的是，这里提出的分类特别关注异构模型之间的推理时间协作。尽管某些机制（例如专家混合 (MoE)）适用于推理和训练阶段，但本次调查将其范围限制在面向推理的场景。也就是说，我们总结并比较了为推理工作负载设计或主要在推理工作负载下评估的协作策略。我们从以下几个角度对现有作品进行分类和回顾：

- 任务分配，将查询路由到最合适的模型（例如，通过基于代理或 MoE 策略）；
- 任务划分，跨模型或系统组件分解执行（例如卸载、请求路由或提前退出机制）；
- 混合策略，使用推测解码和多阶段验证等技术在任务级或令牌级粒度上集成分配和划分。

2.1 任务分配：成本质量权衡的动态模型选择

任务分配决定请求是完全由边缘侧 SLM 还是基于云的 LLM 处理，这与涉及模型间交互的协作解码不同。为了最大限度地减少服务质量 (QoS) 约束下的延迟和能耗，先前的方法采用轻量级评分器、校准奖励估计器或基于老虎机的控制器来在生成之前选择最佳执行路径。一些方法进一步结合用户偏好向量或学习的模型能力表示

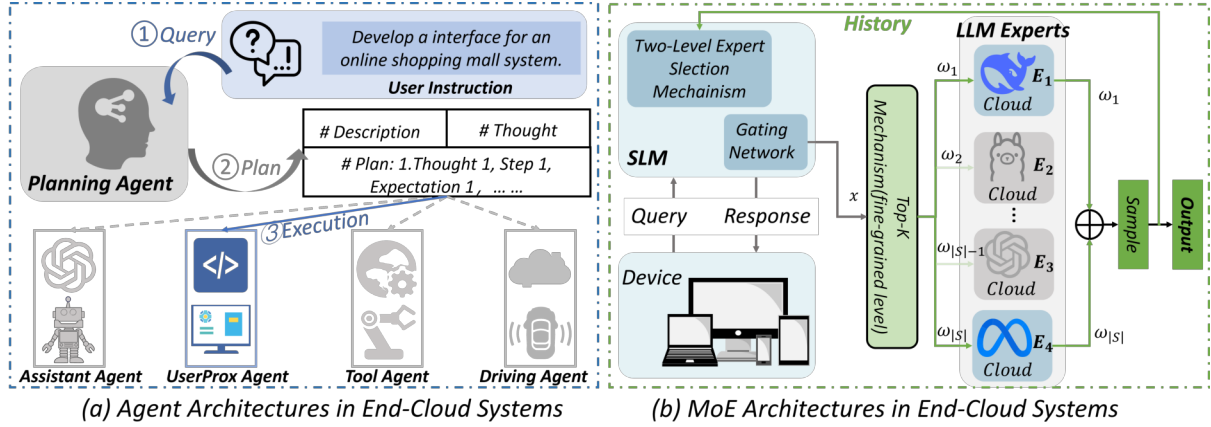


Fig. 3. Comparison of task assignment in edge-cloud systems: (a) Agents decompose user instructions to multi-stage plans executed by specialized agents; (b) MoE-based selection and fusion of top- k cloud LLM experts via fine-grained gating.

adaptive, runtime routing. FS-GEN [84] introduces *collaboration frequency* to quantify the trade-off between large and small models in generation tasks: by aligning input spaces and defining a unified output-fusion objective, FS-GEN measures how often the cloud LLM intervenes in frequency-dominated sequences. Broadly, task assignment strategies fall into three architectural paradigms, resource- and uncertainty-aware assignment, agent-based scheduling and mixture-of-experts (MoE) frameworks (see Fig. 3), each reflecting different coordination granularities and model-selection philosophies.

2.1.1 Resource- and Uncertainty-Aware Task Assignment. To leverage heterogeneous models while respecting edge-cloud capacity constraints (see Table 3), recent work explores diverse task allocation mechanisms that handle runtime uncertainty, system heterogeneity, and model capacity: FS-GEN [34] adopts a dual-system architecture where fast but uncertain SLMs serve as *System 1*, and reliable LLMs act as *System 2*, invoked when ambiguity arises. Fang *et al.* [10] propose a feedback-driven framework that adjusts routing decisions dynamically based on runtime uncertainty, model confidence, and system state, avoiding static thresholds. EdgeLLM [66] uses a value-density-first algorithm to rank tasks by cost-effectiveness, supporting preemption and batched execution via adaptive thresholds. Yang *et al.* [15] introduce fine-grained module-level partitioning, offloading heavy components (*e.g.*, attention, linear layers) to the cloud while retaining lightweight operations at the edge. U-VPA [202] employs uncertainty-guided sampling within a teacher-student framework, selectively uploading domain-informative samples to enhance generalization under non-stationary distributions. A hybrid cost function [123] identifies branch points for edge-cloud transitions, with fault-tolerant re-execution of failed branches locally. Ye *et al.* [204] dynamically select intermediate INT8 representations for edge deployment via automatic tuning. KDSSL [85] leverages LLMs to generate logical rules via UCT-guided search. These rules are verified and deployed on the edge, enabling symbolic reasoning to refine cloud predictions.

2.1.2 Modular Collaboration via MoE and Agent Architectures.

MoE-based approaches. Mixture-of-Experts (MoE) architectures enable scalable, adaptive inference in edge-cloud settings. A lightweight on-device SLM routes queries via a gating or routing network to activate a sparse set of cloud LLM experts (see Fig. 3), whose responses are fused for soft assignment and diversity. MoE² [18] and EdgeMoE [50] guide expert routing using statistical priors or learned policies to reduce memory and I/O overhead. LiteMoE [251] further reduces cost by identifying and merging critical experts without retraining.

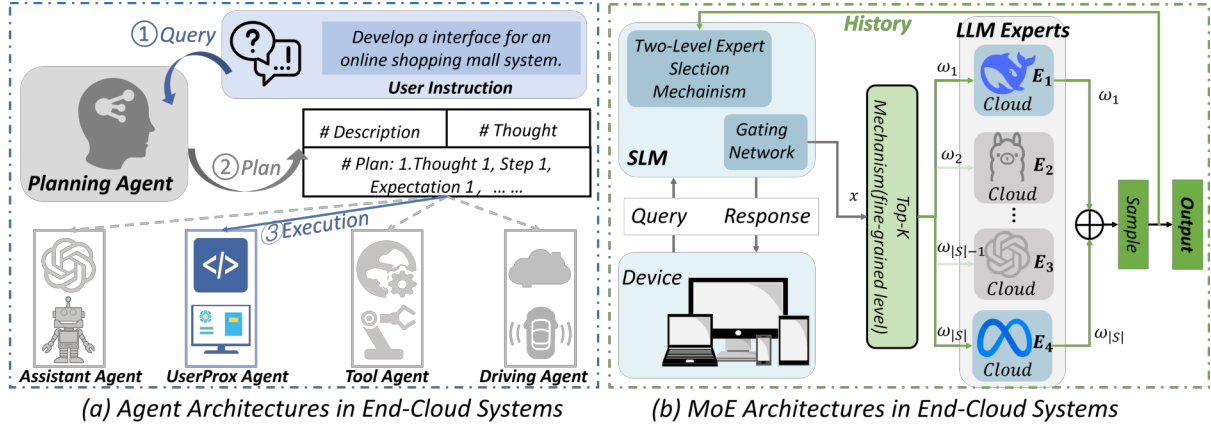


图 3. 边缘云系统中任务分配的比较: (a) 代理将用户指令分解为为专门代理执行的多阶段计划; (b) 通过细粒度选通基于 MoE 的顶级 k 云 LLM 专家的选择和融合。

自适应运行时路由。FS-GEN [84] 引入协作频率来量化生成任务中大小模型之间的权衡: 通过对齐输入空间并定义统一的输出融合目标, FS-GEN 测量云 LLM 干预频率主导序列的频率。从广义上讲, 任务分配策略分为三种架构范式: 资源和不确定性感知分配、基于代理的调度和专家混合 (MoE) 框架 (见图 3), 每种架构范式都反映了不同的协调粒度和模型选择理念。

2.1.1 资源和不确定性感知任务分配。为了利用异构模型, 同时尊重边缘-云容量限制 (见表3), 最近的工作探索了处理运行时不确定性、系统异构性和模型容量的多种任务分配机制: FS-GEN [34] 采用双系统架构, 其中快速但不确定的 SLM 充当系统1, 可靠的 LLM 充当系统2, 在出现歧义时调用。方等人。[10] 提出了一种反馈驱动的框架, 该框架根据运行时不确定性、模型置信度和系统状态动态调整路由决策, 避免静态阈值。EdgeLLM [66] 使用值密度优先算法按成本效益对任务进行排名, 通过自适应阈值支持抢占和批量执行。杨等人。[15] 引入细粒度的模块级分区, 将重型组件 (例如注意力、线性层) 卸载到云端, 同时保留边缘的轻量级操作。U-VPA [202] 在师生框架内采用不确定性引导采样, 有选择地上传领域信息样本以增强非平稳分布下的泛化能力。混合成本函数 [123] 识别边缘-云转换的分支点, 并在本地容错地重新执行失败的分支。叶等人。[204] 通过自动调整动态选择用于边缘部署的中间 INT8 表示。KD SL [85] 利用 LLM 通过 UCT 引导的搜索生成逻辑规则。这些规则在边缘进行验证和部署, 支持符号推理来完善云预测。

2.1.2 通过 MoE 和代理架构进行模块化协作。

基于教育部的的方法。专家混合 (MoE) 架构可在边缘云设置中实现可扩展、自适应推理。轻量级设备上 SLM 通过门控或路由网络路由查询, 以激活一组稀疏的云 LLM 专家 (见图 3), 其响应被融合以实现软分配和多样性。MoE² [18] 和 EdgeMoE [50] 使用统计先验或学习策略来指导专家路由, 以减少内存和 I/O 开销。LiteMoE [251] 通过识别和合并关键专家而无需重新培训, 进一步降低了成本。

Table 3. Resource- and uncertainty-aware task assignment strategies

Reference	Key Idea	Advantage	Limitation
PerLLM [115]	Personalized scheduling with constraint satisfaction	Jointly optimizes deployment and resource usage	Complex constraint modeling
EdgeLLM [66]	Value-density-first scheduling with preemption and batching	Cost-effective, real-time task prioritization	Preemption can destabilize under dynamic loads
FS-GEN [34]	Dual-system inference with uncertainty-triggered LLM fallback	Latency-efficient by conditional large-model invocation	Accuracy depends on uncertainty-estimation precision
Yang <i>et al.</i> [15]	Operator-level scheduling based on compute intensity	Fine-grained control of module placement	Requires precise operator profiling
Stammmer <i>et al.</i> [123]	Hybrid cost-aware branching with fault-tolerant rollback	Reduces re-execution cost during cloud failures	Overhead in maintaining state consistency
U-VPA [202]	Uncertainty-guided sampling in a teacher-student setup	Bandwidth-efficient selection of informative samples	Needs robust domain-shift detection
Li <i>et al.</i> [125, 204]	INT8 model partitioning with automatic tuning	Efficient, low-latency dynamic deployment	Precision loss under aggressive quantization
KDSL [85]	LLM-based rule generation with feedback verification	Closed-loop correction with minimal cloud calls	Complexity in evaluating rule quality

Extending selection to structuring, DoT [49] uses a task decomposition module and dependency-graph scheduler to partition complex queries into subtasks, enabling adaptive granularity and critical-path prioritization. Tian *et al.* [229] propose role-aware MoE routing for multi-turn dialogue, assigning segments to specialized experts and fusing outputs for contextually rich generation. CoEL [272] adds resource-adaptive coordination across edge devices, supporting elastic deployment and efficient scaling under varying budgets.

Agent-based methods. These methods use modular execution pipelines in which a planning agent interprets instructions, formulates structured plans, and delegates subtasks to specialized agents [283] (e.g., assistants, tools, drivers) for context-aware, goal-driven coordination [275]. ARAG [274] integrates four agents, user understanding, natural language inference (NLI), context summarization, and item ranking, into a RAG pipeline. AgentVerse [95] deploys an edge-based coordinator to assemble cloud expert teams by task intent, supporting horizontal aggregation (e.g., voting) and vertical communication for complex reasoning. Salve *et al.* [198] propose a specialized multi-agent framework with a central module orchestrating context-aware query generation and multi-source retrieval. In physically grounded scenarios, WebAgent [230] fuses LLMs with an embodied VirtualHome agent for goal-directed exploration and physical reasoning. ChatEval [228] introduces a debate-style framework where LLM-based judges collaboratively assess generation quality. EcoAgent [254] implements a closed-loop system with a cloud planner and two edge agents, one for execution and one for result verification, that triggers replanning upon failure. Finally, the MADRL framework [257] enables centralized cloud training and decentralized edge execution, reducing training overhead and inference latency in multi-agent reinforcement learning [255].

表 3. 资源和不确定性感知任务分配策略

Reference	Key Idea	Advantage	Limitation
PerLLM [115]	Personalized scheduling with constraint satisfaction	Jointly optimizes deployment and resource usage	Complex constraint modeling
EdgeLLM [66]	Value-density-first scheduling with preemption and batching	Cost-effective, real-time task prioritization	Preemption can destabilize under dynamic loads
FS-GEN [34]	Dual-system inference with uncertainty-triggered LLM fallback	Latency-efficient by conditional large-model invocation	Accuracy depends on uncertainty-estimation precision
Yang <i>et al.</i> [15]	Operator-level scheduling based on compute intensity	Fine-grained control of module placement	Requires precise operator profiling
Stammiller <i>et al.</i> [123]	Hybrid cost-aware branching with fault-tolerant rollback	Reduces re-execution cost during cloud failures	Overhead in maintaining state consistency
U-VPA [202]	Uncertainty-guided sampling in a teacher-student setup	Bandwidth-efficient selection of informative samples	Needs robust domain-shift detection
Li <i>et al.</i> [125, 204]	INT8 model partitioning with automatic tuning	Efficient, low-latency dynamic deployment	Precision loss under aggressive quantization
KDSL [85]	LLM-based rule generation with feedback verification	Closed-loop correction with minimal cloud calls	Complexity in evaluating rule quality

将选择扩展到结构化，DoT [49] 使用任务分解模块和依赖图调度程序将复杂查询划分为子任务，从而实现自适应粒度和关键路径优先级。田等人。[229]提出了用于多轮对话的角色感知 MoE 路由，将片段分配给专业专家并融合输出以实现上下文丰富的生成。CoEL [272]增加了跨边缘设备的资源自适应协调，支持不同预算下的弹性部署和高效扩展。

基于代理的方法。这些方法使用模块化执行管道，其中规划代理解释指令、制定结构化计划并将子任务委托给专门代理[283]（例如助手、工具、驱动程序），以实现上下文感知、目标驱动的协调[275]。A RAG [274] 将四个代理、用户理解、自然语言推理（NLI）、上下文摘要和项目排名集成到 RAG 管道中。AgentVerse [95]部署基于边缘的协调器，根据任务意图组建云专家团队，支持水平聚合（例如投票）和垂直通信以进行复杂推理。萨尔夫等人。[198]提出了一个专门的多代理框架，其中一个中央模块协调上下文感知查询生成和多源检索。在物理接地场景中，WebAgent [230] 将 LLM 与具体的 VirtualHome 代理融合，以进行目标导向的探索和物理推理。ChatEval [228] 引入了一种辩论式框架，其中大语言模型法官可以协作评估生成质量。EcoAgent [254]实现了一个带有云规划器和两个边缘代理的闭环系统，一个用于执行，一个用于结果验证，在失败时触发重新规划。最后，MADRL 框架 [257] 支持集中式云训练和分散式边缘执行，减少多智能体强化学习 [255] 中的训练开销和推理延迟。

Table 4. Comparison of collaborative inference paradigms: routing, offloading, early exit, and communication optimization

Paradigm	Core Innovation	Typical Scenario	Limitations
Routing and Forwarding § 2.2.1	Combines confidence, semantic planning, and contextual bandits for query routing.	Budget- or latency-constrained query routing.	Depends on accurate confidence estimates; fallback may add overhead or misroute.
Computation Offloading § 2.2.2	Supports dynamic scheduling and token-/layer-wise partitioning of models.	Latency-sensitive tasks under fluctuating resource availability.	Performance degrades in highly dynamic or non-stationary environments.
Early Exit § 2.2.3	Enables token- or layer-level early termination via dropout and pipeline reuse.	Low-latency generation or streaming applications.	Accuracy can drop at exit points; modeling token-dependent dynamics is challenging.
Communication Optimization § 2.2.4	Uses entropy compression, bandit routing, and adaptive switching to reduce transfer.	Resource-constrained settings with frequent edge-cloud interactions.	Balancing compression and semantic fidelity remains difficult; sensitive to distribution shifts.

2.2 Task Division: Cooperative Subtask Decomposition Between Large and Small Models

Task division breaks hierarchical or modular tasks into semantic or functional parts, enabling LLMs and SLMs to jointly execute complementary subtasks. This cooperation improves parallelism and responsiveness under edge constraints. Unlike the categorization [287], we identify three fundamentally distinct paradigms of task division in edge-cloud collaborative inference, routing, offloading, and early exit, as illustrated in Fig. 4. Each paradigm offers unique advantages and is suited to different application scenarios, as summarized in Table 4.

2.2.1 Routing and Forwarding Mechanisms. Routing-based methods dynamically select models at inference time to balance latency, cost, and accuracy:

Trust- and Semantic-Aware Routing. These approaches use runtime confidence estimation and semantic planning for adaptive model selection. FrugalGPT [167] reduces API costs via prompt compression and cascaded routing. Tabi [168] employs calibrated confidence scores to choose between lightweight models and powerful LLMs. Dekoninck *et al.* [268] integrate pre- and post-inference quality estimation under a unified framework, improving routing accuracy and fallback success. Kag *et al.* [53, 205] introduce backup-block preloading for robust failover across heterogeneous devices. RouteLLM [106] organizes models hierarchically and applies dynamic programming to minimize tiered performance disparity.

Reward- and Cost-Aware Bandit Routing. These methods use online learning to optimize routing based on reward signals and cost-performance trade-offs. HybridLLM [97] defines a quality gap and uses a lightweight encoder to predict routing probabilities. ZOOPER [166] leverages query-level rewards for expert selection. RouterDC [94] uses dual contrastive loss to align queries with high-performing models. LLM Bandit [269] introduces “model identity vectors” for preference-conditioned routing, while MixLLM [169] and CITER [183] employ contextual bandits to estimate cost and quality. RouteT2I [165] selects between edge and cloud models for text-to-image tasks using multi-dimensional quality metrics.

表 4. 协作推理范式的比较：路由、卸载、提前退出和通信优化

Paradigm	Core Innovation	Typical Scenario	Limitations
Routing and Forwarding § 2.2.1	Combines confidence, semantic planning, and contextual bandits for query routing.	Budget- or latency-constrained query routing.	Depends on accurate confidence estimates; fallback may add overhead or misroute.
Computation Offloading § 2.2.2	Supports dynamic scheduling and token-/layer-wise partitioning of models.	Latency-sensitive tasks under fluctuating resource availability.	Performance degrades in highly dynamic or non-stationary environments.
Early Exit § 2.2.3	Enables token- or layer-level early termination via dropout and pipeline reuse.	Low-latency generation or streaming applications.	Accuracy can drop at exit points; modeling token-dependent dynamics is challenging.
Communication Optimization § 2.2.4	Uses entropy compression, bandit routing, and adaptive switching to reduce transfer.	Resource-constrained settings with frequent edge-cloud interactions.	Balancing compression and semantic fidelity remains difficult; sensitive to distribution shifts.

2.2 任务划分：大小模型之间的协作子任务分解

任务划分将分层或模块化任务分解为语义或功能部分，使 LLM 和 SLM 能够联合执行互补的子任务。这种合作提高了边缘约束下的并行性和响应能力。与分类[287]不同，我们确定了边云协同推理、路由、卸载和提前退出中任务划分的三种根本不同的范式，如图4所示。每种范式都具有独特的优势，适合不同的应用场景，如表4所示。

2.2.1 路由和转发机制。基于路由的方法在推理时动态选择模型，以平衡延迟、成本和准确性：

信任和语义感知路由。这些方法使用运行时置信度估计和语义规划来进行自适应模型选择。FrugalGP T [167] 通过即时压缩和级联路由降低 API 成本。Tabi [168] 使用校准的置信度分数在轻量级模型和强大的大语言模型之间进行选择。德科宁克等人。[268]在统一框架下集成推理前和推理后质量估计，提高路由准确性和回退成功率。卡格等人。[53, 205]引入备份块预加载，以实现跨异构设备的稳健故障转移。RouteLLM [106] 分层组织模型并应用动态编程来最小化分层性能差异。

奖励和成本感知强盗路由。这些方法使用在线学习根据奖励信号和成本性能权衡来优化路由。Hybrid LLM [97]定义了质量差距并使用轻量级编码器来预测路由概率。ZOOTER [166]利用查询级奖励来选择专家。RouterDC [94] 使用双重对比损失来使查询与高性能模型保持一致。LLM Bandit [269] 引入了用于偏好条件路由的“模型身份向量”，而 MixLLM [169] 和 CITER [183] 使用上下文 bandit 来估计成本和质量。RouteT2I [165] 使用多维质量指标在边缘模型和云模型之间选择文本到图像任务。

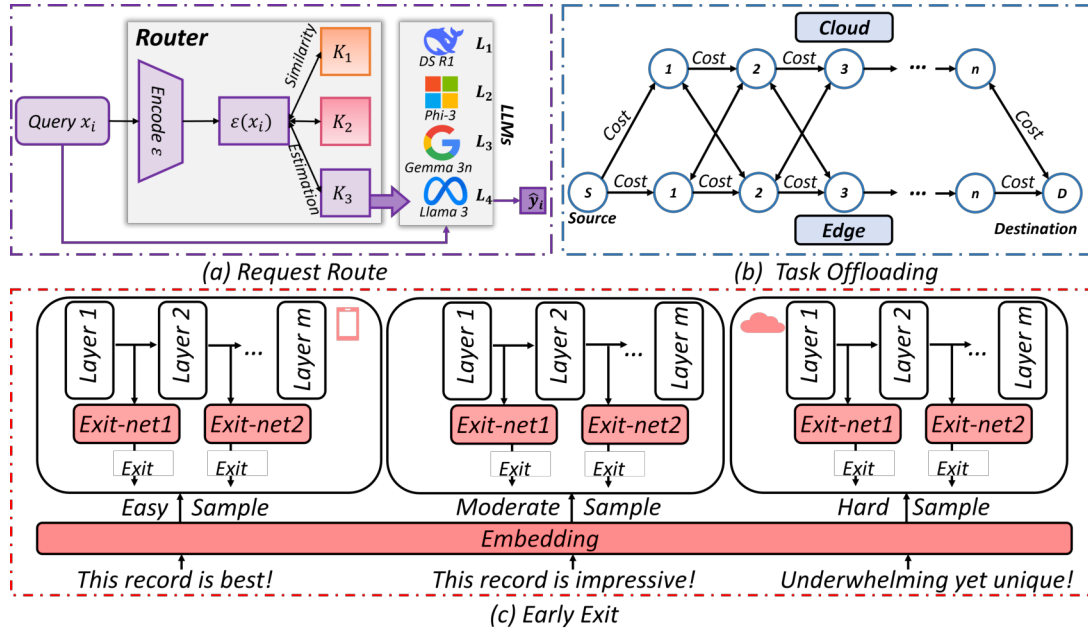


Fig. 4. Illustration of three execution-phase task-division strategies: (a) request routing, dispatching queries to optimal LLM experts based on semantics and cost; (b) task offloading, splitting computation graphs across edge and cloud for cost-aware load balancing; and (c) early exit, multi-exit networks that terminate early for simple inputs to reduce latency and energy.

2.2.2 Computation Offloading and System-Level Optimization. Offloading divides inference tasks across the device and cloud based on runtime conditions: rather than statically assigning tasks from the outset (as in routing), offloading makes dynamic, stage-wise decisions during execution, evaluating current workload against system capacity to maximize performance and utilize resources cost-effectively. We categorize offloading into two complementary types:

Structural Model Partitioning. This strategy decomposes inference by structurally partitioning models across device and edge/cloud at selected layers or token stages according to computational cost and depth [292]. ADAS [55] leverages internet of things (IoT) networks and proposes an enhanced task offloading algorithm based on DDPG, where a diffusion model is employed to generate noise and determine the optimal execution location for each task (i.e., cloud, edge, or local). Liu *et al.* [51] assign lower layers to local devices and offload higher reasoning to edge servers. CE-CoLLM [13] uses token-level confidence to decide which tokens to process locally. Li *et al.* [125] automatically select INT8-quantized intermediate layers as partition points.

Runtime-Adaptive Scheduling. This group of methods dynamically schedules offloading decisions by adapting to real-time metrics, such as latency, confidence, and resource availability, and distributes tasks across local devices, control units, and auxiliary vehicles. Hao *et al.* [12, 124] use device metrics and confidence scores for fine-grained control. He *et al.* [52] introduce a reward-free policy based on latent states. Enhanced Hybrid Inference [65] incorporates user-aware utility models under bandwidth constraints. AVA [132] combines federated and multi-agent reinforcement learning for multi-tier distribution. While these approaches excel at adaptive, condition-aware scheduling, they often struggle to generalize in highly dynamic or heterogeneous environments.

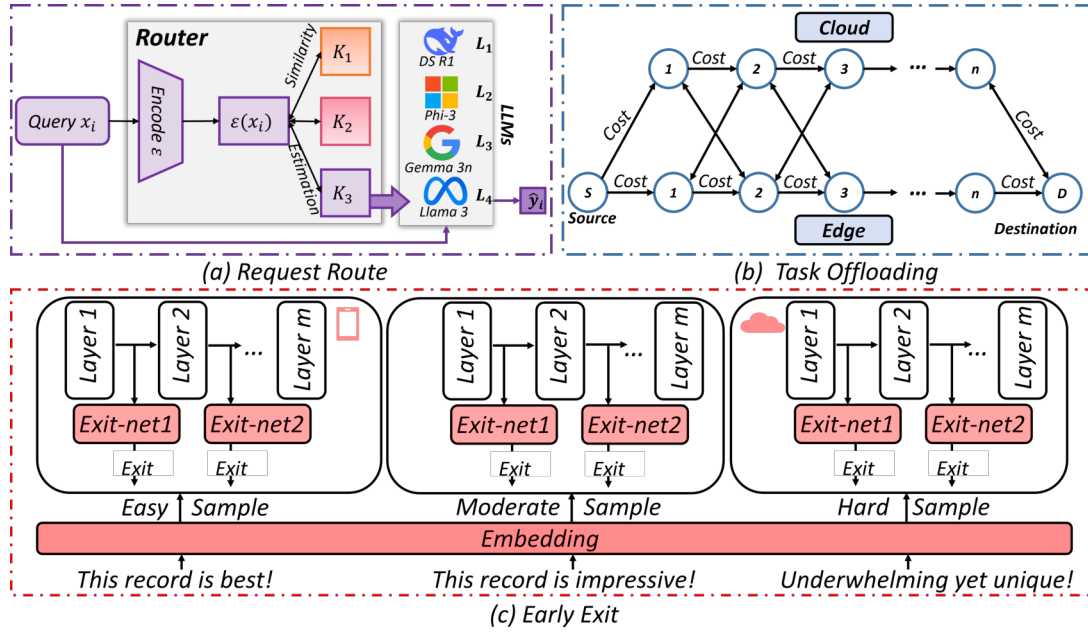


图 4. 三种执行阶段任务划分策略的图示：(a) 请求路由，根据语义和成本将查询分派给最佳 LLM 专家；(b) 任务卸载，跨边缘和云分割计算图以实现成本感知的负载平衡；(c) 提前退出、多退出网络，提前终止简单输入以减少延迟和能量。

2.2.2 计算卸载和系统级优化。卸载根据运行时条件在设备和云之间划分推理任务：卸载不是从一开始就静态分配任务（如在路由中），而是在执行过程中做出动态的分阶段决策，根据系统容量评估当前工作负载，以最大限度地提高性能并经济高效地利用资源。我们将卸载分为两种互补的类型：

结构模型划分。该策略根据计算成本和深度，通过在选定的层或令牌阶段跨设备和边缘/云对模型进行结构性分区来分解推理[292]。ADAS [55]利用物联网（IoT）网络，提出了一种基于 DDPG 的增强型任务卸载算法，其中采用扩散模型来生成噪声并确定每个任务的最佳执行位置（即云、边缘或本地）。刘等人。[51]将较低层分配给本地设备，并将更高层的推理卸载到边缘服务器。CE-CoLLM [13] 使用令牌级别的置信度来决定本地处理哪些令牌。李等人。[125]自动选择INT8量化的中间层作为分区点。

运行时自适应调度。这组方法通过适应实时指标（例如延迟、置信度和资源可用性）动态调度卸载决策，并跨本地设备、控制单元和辅助车辆分配任务。郝等人。[12, 124] 使用设备指标和置信度分数进行细粒度控制。他等人。[52]引入基于潜在状态的无奖励政策。增强型混合推理[65]在带宽限制下结合了用户感知的实用模型。AVA [132] 结合了联邦和多智能体强化学习以实现多层分发。虽然这些方法擅长自适应、条件感知调度，但它们通常难以在高度动态或异构环境中推广。

2.2.3 Early Exit. Early-exit is a hybrid inference paradigm in which execution can terminate at intermediate layers based on system load or model confidence, reducing computation through dynamic exit decisions. LITE [259] introduces a confidence-guided early-exit strategy that halts inference without sacrificing generation quality. LayersKip [130] applies progressively increasing layer dropout and an early-exit loss across Transformer layers to support reliable early termination. EE-LLM [261], designed for 3D-parallelized LLMs, integrates token-level exits via key-value recomputation and pipeline parallelism [252], making it compatible with both large-scale training and inference. EESD [62] uses the initial N layers of a base model and appends a single-layer Transformer to efficiently generate high-quality draft tokens. FREE [258] introduces temporally coherent parallelism by synchronizing shallow processing units with precomputed residuals, accelerating token generation with minimal depth. Collectively, these methods reduce latency and cost but still struggle to maintain consistent accuracy across exit points and coordinate exits under token-dependent dynamics.

Efficient communication is critical for collaborative inference in edge-cloud systems, enabling fine-grained reasoning through joint task division and communication-aware scheduling. Hu *et al.* [267] propose a hybrid architecture that dynamically delegates inference between lightweight edge models and powerful cloud LLMs, achieving 91% diagnostic accuracy with a 28.6% reduction in energy consumption. LLMCascades [111] uses a multi-stage framework in which edge predictions are either accepted or escalated to cloud models via voting and verification, enhancing trustworthiness with minimal redundancy. EdgeShard [117] reduces unnecessary cloud queries by forwarding only inference-critical token features, while entropy-based compression [17] further minimizes transmission cost without accuracy loss. For throughput optimization, PipeEdge [116] partitions LLMs across devices using pipeline parallelism and automated scheduling, reducing latency while preserving model capacity. PerLLM [115] frames coordination as a constrained multi-armed bandit problem, employing an upper confidence bound algorithm to adaptively select the most cost-effective execution path. Blending [271] enables turn-wise model switching in multi-turn interactions, leveraging model heterogeneity to improve response quality while maintaining cost efficiency. Despite these advances, minimizing overhead in dynamic environments and preserving semantic coherence across partitioned reasoning remain open challenges.

2.2.4 Communication Optimization.

2.3 Mixture: Task-Level Orchestration and Delegation in LLM-SLM Collaborative Inference

The *mixture paradigm* hybrids task assignment and execution division, allowing SLMs and LLMs to cooperatively handle a request through staged responsibility sharing. As shown in Fig. 5, we distinguish two granularities: 1) *task-level mixture*, 2) *token-level mixture*, where generation is shared step-by-step across models [129, 196, 203].

2.3.1 Task-Level Decomposition, Scheduling, and Orchestration. Task-level mixture uses semantic cues to identify subtask boundaries and assign them to edge or cloud models based on computational requirements, yielding interpretable, modular workflows. For example, MinionS [138] pipelines edge-side decomposition with cloud aggregation, while HybridSD [7] offloads high-level structural reasoning to the cloud and refines perceptual details on-device. IntellectReq [75] generates abstract intents in the cloud for lightweight edge execution. BAIM [64] integrates multiple edge-model outputs via a gating network for multi-modal collaboration, and OT-GAH [118] uses online tree pruning and assignment heuristics to maximize batch throughput. To streamline chain-of-thought reasoning, HAWKEYE [284] applies length-penalized reinforcement learning in the LLM and offloads detailed expansions to the SLM.

2.3.2 Historical-Enhancement Collaborative Inference. Users' queries to cloud APIs often mirror those sent to local models, motivating methods that leverage historical user-cloud interactions to improve on-device inference. Existing approaches include retrieval-augmented generation, cache-based scheduling, and collaborative learning. For instance, SlimPLM [174] uses heuristic confidence scores to trigger multi-stage retrieval only when the local

2.2.3 提前退出。提前退出是一种混合推理范例，其中执行可以根据系统负载或模型置信度在中间层终止，从而通过动态退出决策减少计算。LITE [259] 引入了一种以置信为导向的提前退出策略，可以在不牺牲生成质量的情况下停止推理。LayersKip [130] 在 Transformer 层上应用逐渐增加的层丢失和提前退出损失，以支持可靠的提前终止。EE-LLM [261] 专为 3D 并行化 LLM 设计，通过键值重新计算和管道并行性 [252] 集成令牌级退出，使其与大规模训练和推理兼容。EESD [62] 使用基本模型的初始 N 层并附加单层 Transformer 来有效生成高质量的草案令牌。FREE [258] 通过将浅层处理单元与预先计算的残差同步来引入时间相干并行性，以最小的深度加速令牌生成。总的来说，这些方法减少了延迟和成本，但仍然难以在退出点上保持一致的准确性，并在依赖于令牌的动态下协调退出。

高效通信对于边缘云系统中的协作推理至关重要，通过联合任务划分和通信感知调度实现细粒度推理。胡等人。[267]提出了一种混合架构，动态委托轻量级边缘模型和强大的云 LLM 之间的推理，实现 91% 的诊断准确率，同时能耗降低 28.6%。LLMCascades [111] 使用多阶段框架，其中边缘预测要么被接受，要么通过投票和验证升级到云模型，以最小的冗余增强可信度。EdgeShard [117] 通过仅转发推理关键令牌特征来减少不必要的云查询，而基于熵的压缩 [17] 进一步最小化传输成本而不损失准确性。为了优化吞吐量，PipeEdge [116] 使用管道并行性和自动调度跨设备对 LLM 进行分区，在保留模型容量的同时减少延迟。PerLLM [115] 将协调框架视为一个受约束的多臂老虎机问题，采用置信上限算法来自适应地选择最具成本效益的执行路径。混合[271]可以在多轮交互中实现轮次模型切换，利用模型异构性来提高响应质量，同时保持成本效率。尽管取得了这些进步，最大限度地减少动态环境中的开销并保持分区推理之间的语义一致性仍然是一个开放的挑战。

2.2.4 通信优化。

2.3 混合：LLM-SLM协同推理中的任务级编排和委托

混合范式混合了任务分配和执行划分，允许 SLM 和 LLM 通过分阶段责任共享来协作处理请求。如图 5 所示，我们区分了两种粒度：1) 任务级混合，2) 令牌级混合，其中生成在模型之间逐步共享[129,196,203]。

2.3.1 任务级分解、调度和编排。任务级混合使用语义线索来识别子任务边界，并根据计算要求将它们分配给边缘或云模型，从而产生可解释的模块化工作流程。例如，MinionS [138] 通过云聚合进行边缘侧分解，而 HybridSD [7] 将高级结构推理卸载到云并细化设备上的感知细节。IntellectReq [75] 在云中生成抽象意图以实现轻量级边缘执行。BAIM [64] 通过门控网络集成多个边缘模型输出以实现多模式协作，而 OT-GAH [118] 使用在线树修剪和分配启发式方法来最大化批量吞吐量。为了简化思想链推理，HAWKEYE [284] 在 LLM 中应用长度惩罚强化学习，并将详细的扩展卸载到 SLM。

2.3.2 历史增强协作推理。用户对云 API 的查询通常会反映发送到本地模型的查询，从而激发了利用历史用户与云交互来改进设备上推理的方法。现有的方法包括检索增强生成、基于缓存的调度和协作学习。例如，SlimPLM [174] 使用启发式置信度分数，仅当本地

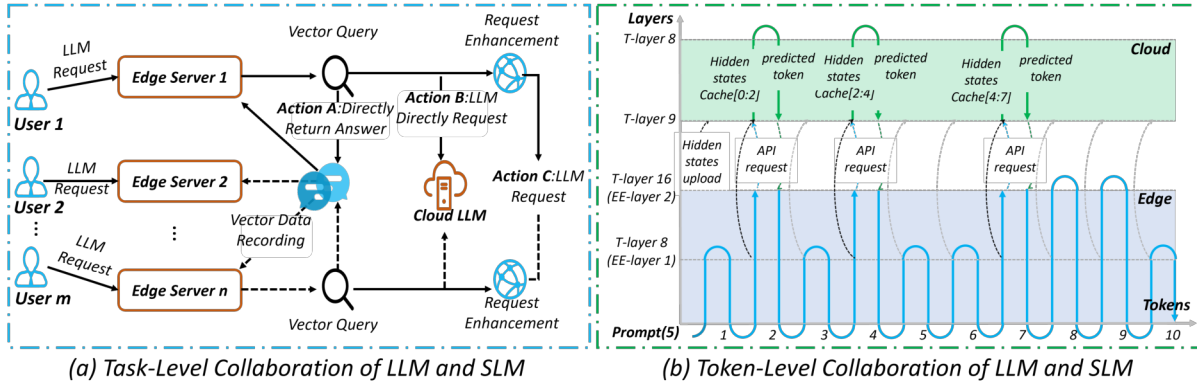


Fig. 5. Mixture strategies for edge-cloud model collaboration: (a) task-level, edge servers decide to respond locally, query the cloud, or forward to LLMs based on context and history; (b) token-level, edge SLMs generate easy tokens locally and offload harder ones to cloud LLMs via hidden-state sharing for fine-grained inference.

response is unreliable. VELO [45] caches prior request embeddings on edge servers and schedules execution paths based on vector similarity. Ding *et al.* [44] store interaction histories for nearest-neighbor retrieval, using subset selection to limit storage overhead. Hybrid-RACA [114] combines a cloud retriever with a lightweight edge predictor by transmitting compressed memory units for enhanced local inference without continuous cloud access. Xu *et al.* [83, 185, 186] introduce an iterative Direct Preference Optimization (DPO) mechanism, where a large model continually guides updates to a small on-device model.

2.3.3 Retrieval-Augmented Generation with Self-Assessment and Feedback. As shown in Fig. 6, these methods leverage external knowledge and reflective reasoning to enhance generation in complex tasks. Qin *et al.* [188, 282] analyze trade-offs among fine-tuning, RAG, data scale, model size, and task difficulty, demonstrating that RAG can outperform fine-tuning under resource constraints and that compressed LLMs benefit more from limited personalized data. Self-Knowledge Retrieval [170] prioritizes internal knowledge, invoking external retrieval only when needed via prompt learning or confidence heuristics. Self-RAG [171] introduces reflection tokens, special control signals that let the model explicitly assess retrieved passages. CRAG [172] employs a lightweight assessor to evaluate retrieval quality and trigger secondary retrieval or web expansion as necessary. Jiang *et al.* [200] extend introspective signals by allowing the model to either incorporate or recalibrate retrieved content. RA-ISF [173] decomposes failed queries into subquestions via a three-stage framework, self-assessment, retrieval, and query decomposition, and integrates subanswers into the final response. SlimRAG [273] builds a lightweight entity-to-fragment index on the cloud for salient entity detection, reducing graph complexity and communication. SpeculativeRAG [211] clusters documents for parallel draft generation.

In practice, Glocker [80] applies timestamp-enhanced RAG in autonomous home object management, enabling robots to track past actions while executing high-level instructions. Zhu *et al.* [190] introduce sparse context selection, parallelizing document encoding and decoding only the most relevant cached content via control tokens. SparseRAG [231] integrates document evaluation and response generation to minimize context loading, accelerating inference and improving output quality for both short- and long-form tasks.

2.4 Mixture: Token-Level Speculation and Verification in Edge-Cloud LLM-SLM Collaboration

Most LLMs generate tokens autoregressively, producing each token sequentially for the next prediction. While this yields high-quality output, it incurs substantial latency, which limits real-time deployment on resource-constrained

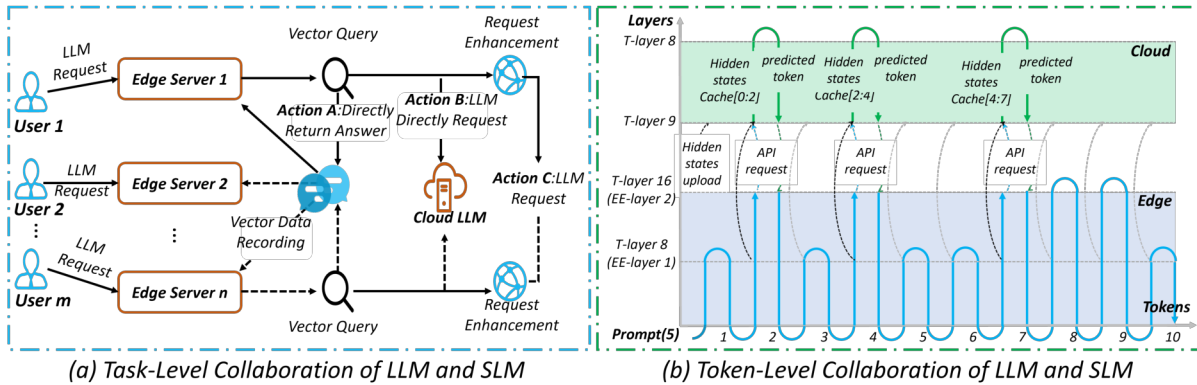


图 5. 边缘-云模型协作的混合策略: (a) 任务级, 边缘服务器根据上下文和历史决定本地响应、查询云或转发到 LLM; (b) 令牌级边缘 SLM 在本地生成简单的令牌, 并通过隐藏状态共享将较难的令牌卸载到云 LLM 以进行细粒度推理。

响应不可靠。VELO [45] 在边缘服务器上缓存先前的请求嵌入, 并根据向量相似性来调度执行路径。丁等人。[44] 存储最近邻检索的交互历史, 使用子集选择来限制存储开销。Hybrid-RACA [114] 通过传输压缩内存单元将云检索器与轻量级边缘预测器结合起来, 以增强本地推理而无需连续的云访问。徐等人。[83,185,186] 引入了迭代直接偏好优化 (DPO) 机制, 其中大型模型不断引导小型设备上模型的更新。

2.3.3 具有自我评估和反馈的检索增强生成。如图 6 所示, 这些方法利用外部知识和反思推理来增强复杂任务的生成。秦等人。[188, 282] 分析了微调、RAG、数据规模、模型大小和任务难度之间的权衡, 证明 RAG 在资源限制下可以优于微调, 并且压缩的大语言模型从有限的个性化数据中受益更多。自知识检索 [170] 优先考虑内部知识, 仅在需要时通过即时学习或置信启发法调用外部检索。Self-RAG [171] 引入了反射标记, 这是一种特殊的控制信号, 可以让模型明确地评估检索到的段落。CRAG [172] 采用轻量级评估器来评估检索质量并根据需要触发二次检索或网络扩展。江等人。[200] 通过允许模型合并或重新校准检索到的内容来扩展内省信号。RA-ISF [173] 通过三阶段框架、自我评估、检索和查询分解将失败的查询分解为子问题, 并将子答案集成到最终响应中。SlimRAG [273] 在云上构建轻量级实体到片段索引, 用于显着实体检测, 降低图形复杂性和通信。SpeculativeRAG [211] 对文档进行聚类以并行草稿生成。

在实践中, Glocker [80] 在自主家居对象管理中应用了时间戳增强型 RAG, 使机器人能够在执行高级指令的同时跟踪过去的动作。朱等人。[190] 引入稀疏上下文选择, 通过控制令牌并行化文档编码和仅解码最相关的缓存内容。SparseRAG [231] 集成了文档评估和响应生成, 以最大限度地减少上下文加载, 加速推理并提高短格式和长格式任务的输出质量。

2.4 混合: 边缘云 LLM-SLM 协作中的代币级推测和验证

大多数大语言模型以自回归方式生成标记, 为下一个预测顺序生成每个标记。虽然这会产生高质量的输出, 但会延迟, 从而限制了资源受限的实时部署

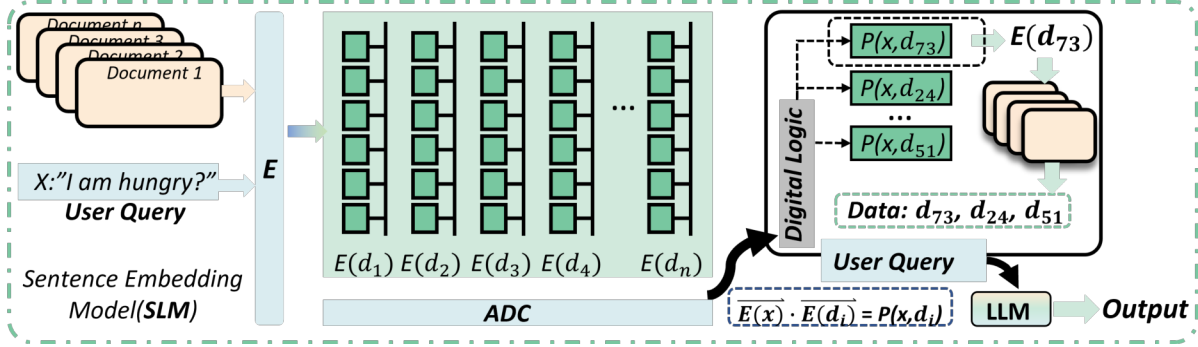


Fig. 6. Hybrid SLM-LLM collaboration in RAG: the SLM encodes queries and retrieves documents via approximate nearest-neighbor search, and the LLM generates responses using the query and retrieved context.

devices. To address this, recent work [244] proposes a “lightweight drafting + precise verification” paradigm: the edge-deployed SLM rapidly generates high-confidence drafts, and the cloud-based LLM verifies semantic consistency, corrects uncertain tokens, and enriches content with additional detail [276]. As shown in Fig. 7, we survey speculative decoding architectures and techniques in edge-cloud collaboration, covering draft-refine frameworks, draft-completion strategies, skeleton-completion mechanisms, and token-level verification [14].

2.4.1 Edge Draft and Cloud Validation. This line of work divides into two categories: 1) vanilla speculative decoding frameworks with algorithmic enhancements and 2) system-level designs that improve parallelism and reduce latency under diverse deployment constraints.

Vanilla Speculative Decoding and Algorithmic Enhancements. Early works such as [32, 40, 208] introduced the basic speculative decoding paradigm. A fallback mechanism [207] lets the lightweight model defer uncertain tokens to the larger model, while a rollback strategy corrects inaccuracies. Building on this, RSD [37] and SpecExec [42] incorporate controlled bias to prioritize high-reward outputs. AutoMix [92] integrates symbolic reasoning with reinforcement learning to mitigate hallucinations [213] and reasoning errors, using kernel density estimation to close the feedback loop between on-device verification and cloud routing. Fu *et al.* [215] improve efficiency by alternating the draft and target models as proposer and verifier via a learned verification distribution.

Parallel and Low-Latency Speculative Inference Frameworks. Interactive applications (*e.g.*, question answering, voice assistants, real-time translation) demand low Time-To-First-Token (TTFT) and Token-By-Token Time (TBT). DiSCo [41] estimates the acceptance probability of each speculative token using the drafting model’s logits. To avoid mutual stalling, where the draft model waits for validation, SpecDec [100, 109] performs parallel token verification with soft discrimination. PEARL [191] uses adaptive draft lengths and early verification to accelerate decoding, and SEED [210] manages drafts with a round-robin FCFS queue.

Multi-Layer and Lightweight Verification Pipelines. To reduce cloud load, [135, 195, 219, 223] propose multi-layer pipelines: a lightweight verifier filters token blocks before final cloud validation [217]. Falcon [194] boosts intra-block token dependency via coupled sequential scanning, improving speculative accuracy while maintaining a compact two-layer transformer draft model suited for lightweight, multi-layer verification. Spector [128] adds an extra draft stage using a simpler model to pre-filter low-confidence hypotheses. [291] uses layer parallelism via early exits for mid-verification token drafting. Clients pre-draft tokens using idle time before final validation, boosting edge-cloud parallelism. This achieves a 21% speedup over cloud decoding on the Unitree Go 2 robot.

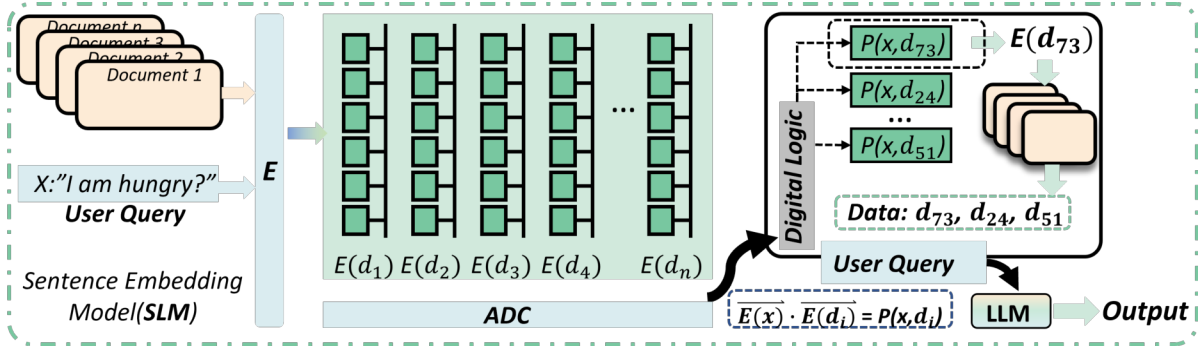


图 6. RAG 中的混合 SLM-LLM 协作：SLM 通过近似最近邻搜索对查询进行编码并检索文档，LLM 使用查询和检索到的上下文生成响应。

设备。为了解决这个问题，最近的工作[244]提出了一种“轻量级起草+精确验证”范例：边缘部署的SLM快速生成高置信度的草稿，基于云的LLM验证语义一致性，纠正不确定的标记，并用额外的细节丰富内容[276]。如图7所示，我们调查了边缘-云协作中的推测性解码架构和技术，涵盖草稿-细化框架、草稿-完成策略、骨架-完成机制和令牌级验证[14]。

2.4.1 边缘草稿和云验证。该工作线分为两类：1) 具有算法增强功能的普通推测解码框架，2) 在不同部署约束下提高并行性并减少延迟的系统级设计。

普通推测解码和算法增强。早期的作品如[32,40,208]介绍了基本的推测解码范式。回退机制[207]让轻量级模型将不确定的标记推迟到更大的模型，而回滚策略则纠正不准确。在此基础上，RSD [37] 和 SpecExec [42] 结合了受控偏差来优先考虑高回报输出。AutoMix [92] 将符号推理与强化学习相结合，以减轻幻觉 [213] 和推理错误，使用内核密度估计来关闭设备上验证和云路由之间的反馈循环。傅等人。[215]通过学习的验证分布将草稿模型和目标模型交替作为提议者和验证者来提高效率。

并行和低延迟推测推理框架。交互式应用程序（例如问答、语音助手、实时翻译）需要较低的首次令牌时间 (TTFT) 和逐令牌时间 (TBT)。DiSCo [41] 使用起草模型的 logits 估计每个投机令牌的接受概率。为了避免草稿模型等待验证的相互停滞，SpecDec [100, 109] 使用软区分执行并行令牌验证。PEARL [191] 使用自适应草稿长度和早期验证来加速解码，SEED [210] 使用循环 FCFS 队列管理草稿。

多层和轻量级验证管道。为了减少云负载，[135,195,219,223]提出了多层管道：轻量级验证器在最终云验证之前过滤令牌块[217]。Falcon [194] 通过耦合顺序扫描增强了块内令牌依赖性，提高了推测准确性，同时保持了适合轻量级多层验证的紧凑两层变压器草案模型。Spector [128] 使用更简单的模型添加了一个额外的草案阶段来预过滤低置信度假设。[291]通过提前退出来使用层并行性来进行中期验证令牌起草。客户在最终验证之前利用空闲时间预先起草令牌，从而提高边缘云并行性。与 Unitree Go 2 机器人上的云解码相比，这实现了 21% 的加速。

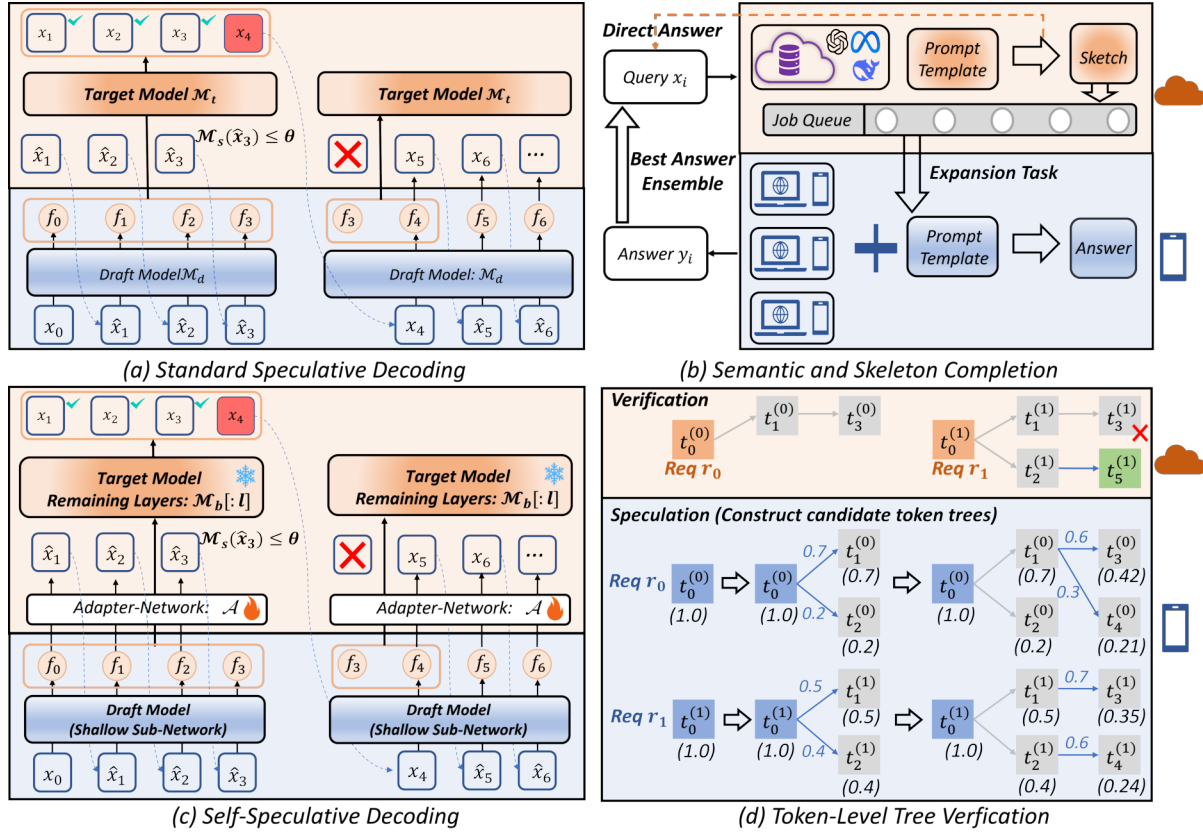


Fig. 7. (a) Vanilla: the cloud verifies edge-generated drafts and rejects or regenerates invalid tokens; (b) Semantic Skeleton: the cloud generates high-level semantic skeletons via prompts, which the edge completes with identical or refined prompts; (c) Self-Speculative: the model is split into shallow and deep subnetworks, using adapters for lightweight on-device verification; (d) Token Tree: token trees are expanded probabilistically, and the cloud verifies optimal branches per request.

Draft Quality Enhancement and Hardware-Aware Scheduling. Fuzzy Speculative Decoding (FSD) [27] relaxes strict match constraints, allowing minor mismatches to boost speed [221]. Zhao *et al.* [126, 136] propose phrase-level reuse, caching validated phrases for rapid generation. DistillSpec [137] distills the target model into the draft model offline using self-sampled data. DuoDecoding [28] assigns drafting to the CPU and validation to the GPU, dynamically adjusting draft lengths based on system load [218]. SPIN [290] uses a learning-based SSM selection without request priors and accelerates execution via GPU pipelining of speculation and verification. BanditSpec [214] formulated adaptive hyperparameter selection problem when generating text as a multi arm Bandit problem and designed two bandit based hyperparameter selection algorithms.

2.4.2 Self-Speculative Decoding without Auxiliary Draft Models. Some studies [61, 216] note that adding a separate draft model increases communication, storage, and training costs [253]. Kangaroo [38] reuses the target LLM’s shallow subnetwork and LM head for self-drafting, significantly reducing overhead. SWIFT [61] exploits context-aware layer skipping for dynamic computation, and ASD [216] extends speculative sampling to diffusion models

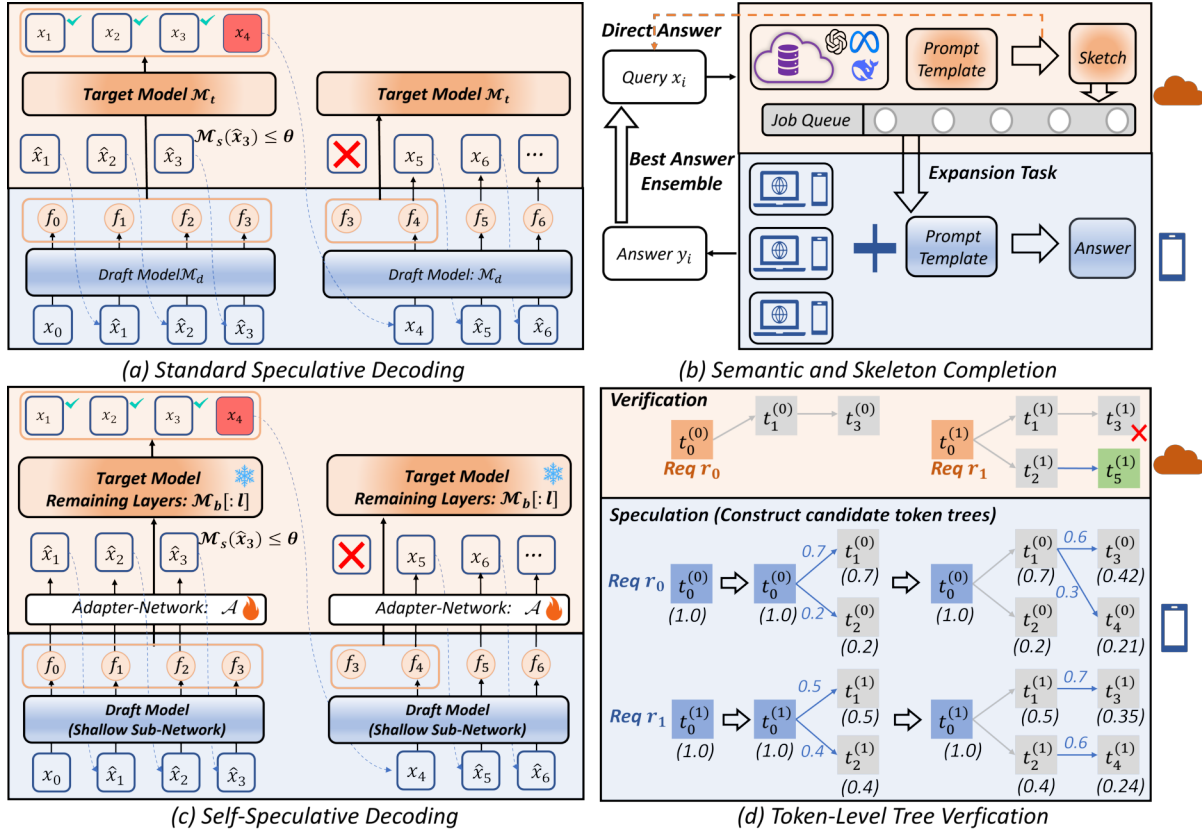


图 7. (a) Vanilla: 云验证边缘生成的草稿并拒绝或重新生成无效令牌; (b) 语义骨架: 云通过提示生成高级语义骨架, 边缘用相同或细化的提示完成该骨架; (c) 自推测: 模型分为浅层子网和深层子网, 使用适配器进行轻量级设备上验证; (d) 令牌树: 令牌树按概率扩展, 云验证每个请求的最佳分支。

草案质量增强和硬件感知调度。模糊推测解码 (FSD) [27] 放宽了严格的匹配约束, 允许较小的不匹配来提高速度[221]。赵等人。[126, 136] 提出短语级重用, 缓存经过验证的短语以快速生成。DistillSpec [137] 使用自采样数据离线将目标模型提炼为草稿模型。DuoDecoding [28] 将绘图分配给 CPU, 将验证分配给 GPU, 根据系统负载动态调整绘图长度 [218]。SPIN [290] 使用基于学习的 SSM 选择, 无需先验请求, 并通过推测和验证的 GPU 流水线加速执行。BanditSpec [214] 将生成文本时的自适应超参数选择问题表述为多臂 Bandit 问题, 并设计了两种基于 bandit 的超参数选择算法。

2.4.2 没有辅助草案模型的自推测解码。一些研究 [61, 216] 指出, 添加单独的草稿模型会增加通信、存储和培训成本 [253]。Kangaroo [38] 重用目标 LLM 的浅层子网络和 LM 头进行自起草, 显着减少了开销。S WIFT [61] 利用上下文感知层跳跃进行动态计算, ASD [216] 将推测采样扩展到扩散模型

Table 5. Comparison of cloud-to-edge and edge-to-cloud skeleton collaboration paradigms

Paradigm	Ref.	Key Ideas	Advantages	Limitations
Cloud-to-Edge	PICE [43]	Progressive inference: cloud LLM drafts, edge SLM refines	Reduces latency and supports incremental, real-time reasoning	Requires fine-grained synchronization between LLM and SLM
	CoGenesis [33]	Sketch planning by LLM with local completion; distribution-only collaboration via logits	Enables privacy-preserving inference and adapts to bandwidth constraints	Coarse plans may limit expressiveness; logit protocols are model-specific
	NEST [201]	Cloud performs full retrieval, edge conducts token-level neighbor search for caching	Reduces corpus storage and improves retrieval latency	Requires careful tuning of retrieval-token matching; introduces multi-stage complexity
Edge-to-Cloud	SlimPLM [174]	Triggers LLM only when edge model uncertainty is high	Minimizes cloud calls and provides fast edge responses	Depends on accurate uncertainty detection; may miss subtle errors
	Hao <i>et al.</i> [14]	Token correction: LLM refines output by editing a few edge tokens	Efficient corrections with minimal cloud usage	Requires reliable error localization; not suited for structural rewrites
	Probe Sampling [209]	Measures draft-target similarity to decide cloud fallback	Lowers cloud overhead while maintaining output fidelity	Relies on precise similarity estimation; risk of under-correction

by leveraging DDPM exchangeability for parallel inference. These methods maintain speed gains without extra model training.

2.4.3 Semantic and Skeleton Completion. The draft-refine paradigm splits generation into two stages: edge-side drafting and cloud-side refinement (see Table 5).

Cloud-to-Edge Skeletons. This class of methods adopts a top-down strategy, where the cloud LLM first produces a high-level semantic “skeleton”, and the edge-deployed SLM complements or adapts the content based on local context or constraints. PICE [43] generates a task sketch for concurrent edge refinement, reducing latency. CoGenesis [33] offers 1) sketch-based planning with local completion and 2) logit-based privacy-preserving inference. NEST [201] uses hybrid retrieval in the cloud and token-level neighbor search on the edge via lightweight caches.

Edge-to-Cloud Drafting. This paradigm starts with fast, low-cost draft generation at the edge, followed by refinement from a more capable cloud model. SlimPLM [174] triggers LLM engagement based on local knowledge sufficiency. Hao *et al.* [14] perform token-level corrections, selectively replacing ambiguous tokens. Probe Sampling [209] filters prompts based on draft-target similarity, minimizing cloud usage. While collaboration granularity varies, from document-chunk retrieval to token-level correction, these methods exemplify two complementary coordination paradigms: cloud-to-edge sketching for semantic decomposition and edge-to-cloud refinement for lightweight, responsive drafting.

表 5 云到边和边到云骨架协作范式比较

Paradigm	Ref.	Key Ideas	Advantages	Limitations
e d g e : c l o u d : e d g e	PICE [43]	Progressive inference: cloud LLM drafts, edge SLM refines	Reduces latency and supports incremental, real-time reasoning	Requires fine-grained synchronization between LLM and SLM
	CoGenesis [33]	Sketch planning by LLM with local completion; distribution-only collaboration via logits	Enables privacy-preserving inference and adapts to bandwidth constraints	Coarse plans may limit expressiveness; logit protocols are model-specific
	NEST [201]	Cloud performs full retrieval, edge conducts token-level neighbor search for caching	Reduces corpus storage and improves retrieval latency	Requires careful tuning of retrieval-token matching; introduces multi-stage complexity
d d p m : e d g e : c l o u d	SlimPLM [174]	Triggers LLM only when edge model uncertainty is high	Minimizes cloud calls and provides fast edge responses	Depends on accurate uncertainty detection; may miss subtle errors
	Hao <i>et al.</i> [14]	Token correction: LLM refines output by editing a few edge tokens	Efficient corrections with minimal cloud usage	Requires reliable error localization; not suited for structural rewrites
	Probe Sampling [209]	Measures draft-target similarity to decide cloud fallback	Lowers cloud overhead while maintaining output fidelity	Relies on precise similarity estimation; risk of under-correction

利用 DDPM 可交换性进行并行推理。这些方法无需额外的模型训练即可保持速度增益。

2.4.3 语义和骨架完成。草稿-细化范式将生成分为两个阶段：边缘草稿和云端细化（参见表 5）。

云到边缘的骨架。此类方法采用自上而下的策略，其中云LLM首先生成高级语义“骨架”，边缘部署的SLM根据本地上下文或约束补充或调整内容。PICE [43] 生成一个用于并发边缘细化的任务草图，从而减少延迟。CoGenesis [33] 提供 1) 基于草图的局部完成规划和 2) 基于 logit 的隐私保护推理。NEST [201] 在云中使用混合检索，并通过轻量级缓存在边缘使用令牌级邻居搜索。

边缘到云绘图。这种范例首先在边缘快速、低成本地生成草稿，然后从功能更强大的云模型中进行细化。SlimPLM [174] 基于本地知识充足性触发大语言模型参与。郝等人。[14]执行令牌级别的更正，有选择地替换不明确的令牌。探测采样[209]根据草稿目标相似性过滤提示，最大限度地减少云的使用。虽然协作粒度各不相同，从文档块检索到令牌级校正，但这些方法体现了两种互补的协调范例：用于语义分解的云到边缘草图和用于轻量级响应式绘图的边缘到云细化。

Table 6. Comparison of collaborative training paradigms in edge-cloud LLM-SLM systems

Category	Definition & Characteristics	Advantages	Limitations
Distillation-Based Collaboration	Uses logits, features, or hidden states for LLM-to-SLM supervision with task- and domain-adaptive schemes	Flexible; enables effective model compression and personalization	Sensitive to domain shift and teacher-student mismatch; data quality-dependent; uni-directional by default
Multi-SLM Parameter Fusion	Integrates heterogeneous SLMs via stitching, subnet assembly and capacity-guided transfer	Enhances edge personalization; enables cross-domain parameter reuse; supports one-pass deployment	Structural mismatch sensitivity; requires functional/meta-guidance; faces stability challenges
Adapter-Based Modular Training	Inserts LoRA-like adapters for parameter-efficient fine-tuning and federated updates	Scalable across tasks and devices; reduces communication costs; preserves base model integrity	Depends on shared adapter architectures; limited in cross-modal or non-aligned scenarios
SLM-Driven LLM Supervision	Employs SLM CoT to refine LLM outputs in noisy/low-resource contexts	Improves interpretability; enhances reliability on simple or repetitive tasks	Constrained by SLM capacity; risk of reinforcing local biases
Cloud-Guided Capability Injection	LLMs generate task-specific modules or behavioral updates for SLMs via meta-learning or distillation	Enables few-shot adaptation; supports evolving edge demands; modular and task-aware	Requires costly cloud computation; may induce concept drift; on-device validation is challenging

2.4.4 Token Tree Verification. In speculative decoding, optimal verification timing is crucial: early verification may waste computation, while delayed checks risk irreversible errors (see Fig. 7). To mitigate this, the token tree structure allows each node to branch into multiple candidate paths, enabling broader output exploration and reducing waste from single-path errors. Along this line, LLMcad [26] and Traversal Verification [133] introduce a non-autoregressive token-tree verifier capable of simultaneously validating and correcting all branches within a single iteration. Building on this foundation, AdaServe [39] constructs a candidate token tree for each request and dynamically selects the optimal token branch to maximize system throughput under predefined service level objectives (SLOs) [29, 131]. OPT-Tree [60] addresses the inefficiency of fixed-structure token trees by proposing a dynamic tree construction method. It greedily builds an expectation-optimal candidate tree at each decoding step and prunes branches via probabilistic modeling, thus maximizing valid tokens under a node budget. Traversal Verification [133] further enhances tree utilization by introducing bottom-up, sequence-level verification. This method evaluates complete token paths rather than individual tokens, reducing the risk of prematurely discarding useful subsequences. Sequoia [127] builds on this by formulating token tree construction as a dynamic programming problem. It avoids draft model resampling and integrates hardware-aware optimizations to adaptively select tree depth and size for target platforms.

3 Overview of Collaborative learning Architectures

Collaborative training facilitates ongoing knowledge exchange between edge SLMs and cloud LLMs under heterogeneous conditions. Unlike latency-focused inference cooperation, training stresses structural compatibility, non-IID adaptation, and privacy-preserving transfer. Traditional distillation and transfer learning assume aligned

表 6. 边缘云 LLM-SLM 系统中协作训练范例的比较

Category	Definition & Characteristics	Advantages	Limitations
Distillation-Based Collaboration	Uses logits, features, or hidden states for LLM-to-SLM supervision with task- and domain-adaptive schemes	Flexible; enables effective model compression and personalization	Sensitive to domain shift and teacher-student mismatch; data quality-dependent; uni-directional by default
Multi-SLM Parameter Fusion	Integrates heterogeneous SLMs via stitching, subnet assembly and capacity-guided transfer	Enhances edge personalization; enables cross-domain parameter reuse; supports one-pass deployment	Structural mismatch sensitivity; requires functional/meta-guidance; faces stability challenges
Adapter-Based Modular Training	Inserts LoRA-like adapters for parameter-efficient finetuning and federated updates	Scalable across tasks and devices; reduces communication costs; preserves base model integrity	Depends on shared adapter architectures; limited in cross-modal or non-aligned scenarios
SLM-Driven LLM Supervision	Employs SLM CoT to refine LLM outputs in noisy/low-resource contexts	Improves interpretability; enhances reliability on simple or repetitive tasks	Constrained by SLM capacity; risk of reinforcing local biases
Cloud-Guided Capability Injection	LLMs generate task-specific modules or behavioral updates for SLMs via meta-learning or distillation	Enables few-shot adaptation; supports evolving edge demands; modular and task-aware	Requires costly cloud computation; may induce concept drift; on-device validation is challenging

2.4.4 令牌树验证。在推测性解码中，最佳验证时机至关重要：早期验证可能会浪费计算，而延迟检查则可能会出现不可逆错误（见图 7）。为了缓解这种情况，令牌树结构允许每个节点分支到多个候选路径，从而实现更广泛的输出探索并减少单路径错误造成的浪费。沿着这条线，LLMCad [26] 和遍历验证 [133] 引入了一种非自回归令牌树验证器，能够在单次迭代中同时验证和纠正所有分支。在此基础上，Ada Serve [39] 为每个请求构建候选令牌树，并动态选择最佳令牌分支，以在预定义的服务级别目标 (SLO) [29, 131] 下最大化系统吞吐量。OPT-Tree [60] 通过提出动态树构建方法来解决固定结构令牌树的低效率问题。它在每个解码步骤贪婪地构建期望最优候选树，并通过概率建模修剪分支，从而在节点预算下最大化有效令牌。遍历验证 [133] 通过引入自下而上的序列级验证进一步增强了树的利用率。该方法评估完整的令牌路径而不是单个令牌，从而降低了过早丢弃有用子序列的风险。Sequoia [127] 在此基础上将令牌树构建表述为动态编程问题。它避免了草稿模型重采样，并集成了硬件感知优化，以自适应地为目标平台选择树深度和大小。

3 协作学习架构概述

协作培训促进边缘 SLM 和云 LLM 在异构条件下持续进行知识交换。与注重延迟的推理合作不同，训练强调结构兼容性、非独立同分布适应和隐私保护传输。传统的蒸馏和迁移学习假设一致

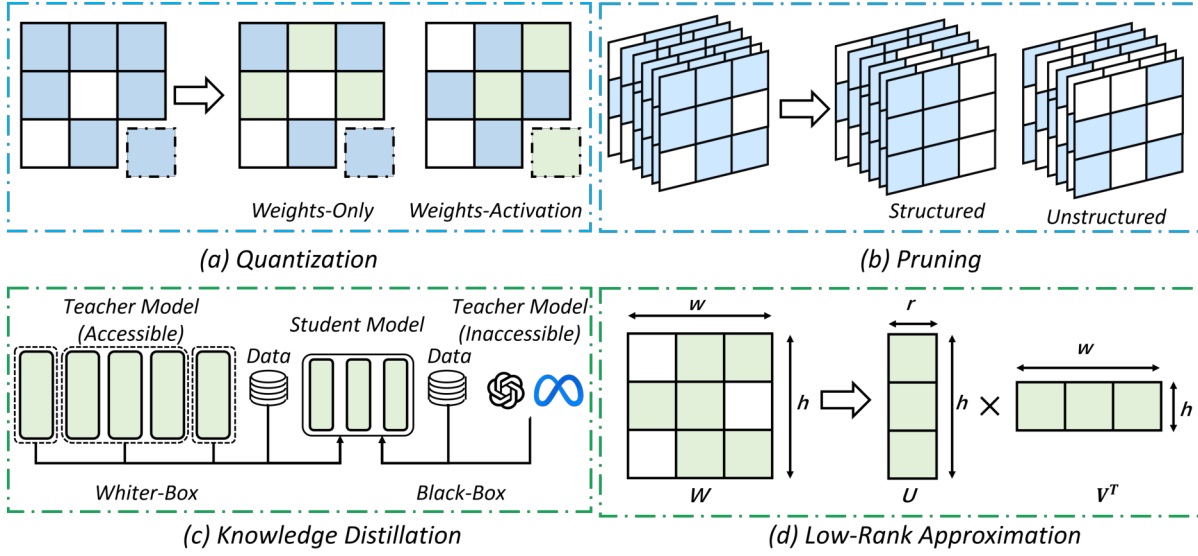


Fig. 8. Model compression techniques for efficient cloud-edge collaborative training: (a) quantization of weights and activations; (b) pruning of parameters (structured or unstructured sparsity); (c) knowledge distillation for teacher-student transfer; and (d) low-rank approximation of weight matrices.

architectures or tasks, limiting generality. In practice, structural gaps and data heterogeneity cause biases and spurious correlations that harm aggregation and generalization. Liu *et al.* [90] formalize large-small model collaboration under privacy, exposure, and cost constraints, decomposing it into three stages: 1) LLM-to-SLM transfer, 2) SLM-to-LLM feedback, and 3) closed-loop adaptation. To unify this landscape, we propose a taxonomy of collaborative training across five paradigms: distillation-based collaboration, adapter-based modular training, bidirectional learning, SLM-driven supervision, and cloud-guided capability injection. As shown in Table 6, our classification highlights each paradigm’s functional role and progressive interdependence, from unidirectional transfer to modular tuning, iterative feedback, and reversed supervision. Grounded in knowledge flow, coupling strength, structural assumptions, and deployment constraints, this taxonomy provides a principled basis for selecting suitable strategies in edge-cloud contexts and inspires more robust, generalizable training designs.

3.1 Pruning and Quantization Strategies for Efficient Device-Cloud Collaboration

In device-cloud collaborative training frameworks, pruning and quantization are key techniques for constructing lightweight models that reconcile edge devices’ limited resources with cloud models’ accuracy demands. Li *et al.* [120] introduce a sparsity-aware channel pruning method that evaluates feature distribution deviations to remove globally unimportant channels; its soft-mask mechanism allows selective reactivation. EfficientLLM [121] applies progressive structural pruning during pretraining to eliminate redundant substructures. Complementing pruning, Jiang *et al.* [122] propose a split-transformer design in which a lightweight edge encoder produces quantized intermediate embeddings for cloud decoding. MergeNet [67] addresses structural incompatibility via low-rank decomposition of LLM and SLM parameters, followed by attention-based fusion, enabling expressive knowledge transfer without altering the edge architecture. As illustrated in Fig. 8, these methods co-optimize pruning and quantization across training, inference, and scheduling dimensions. For task-granularity mixtures in

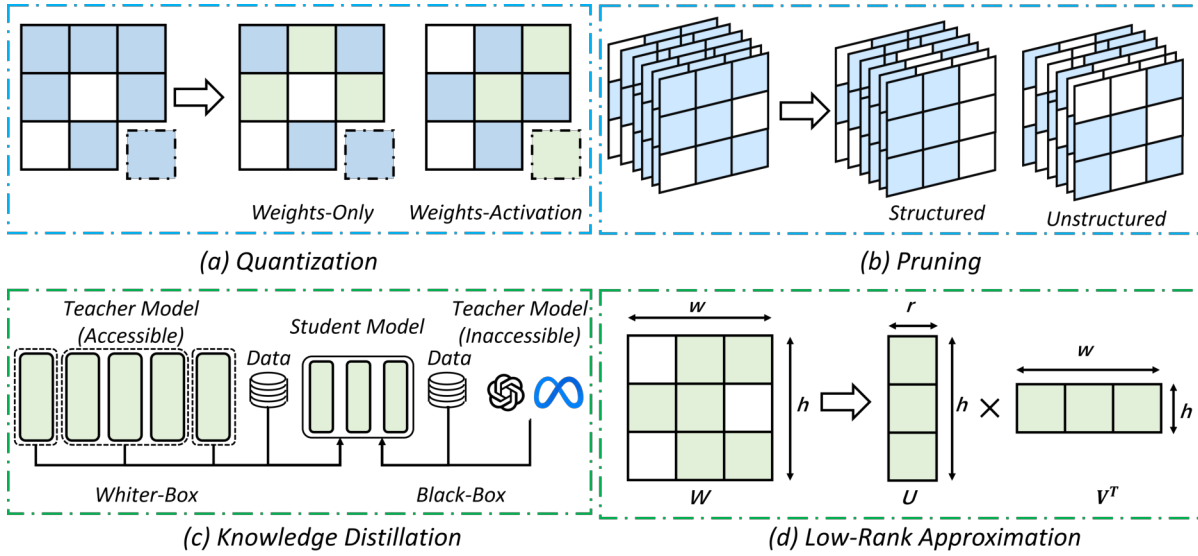


图 8. 高效云边协作训练的模型压缩技术: (a) 权重和激活的量化; (b) 参数修剪 (结构化或非结构化稀疏性); (c) 师生转移的知识提炼; (d) 权重矩阵的低阶近似。

架构或任务, 限制了通用性。在实践中, 结构差距和数据异质性会导致偏差和虚假相关性, 从而损害聚合和泛化。刘等人。[90]在隐私、暴露和成本约束下正式化大小模型协作, 将其分解为三个阶段: 1) LLM到SLM转移, 2) SLM到LLM反馈, 3) 闭环适应。为了统一这一格局, 我们提出了跨五种范式的协作训练分类法: 基于蒸馏的协作、基于适配器的模块化训练、双向学习、SLM 驱动的监督 and 云引导的能力注入。如表 6 所示, 我们的分类突出了每个范式的功能作用和渐进的相互依赖性, 从单向转移到模块化调整、迭代反馈和反向监督。该分类法以知识流、耦合强度、结构假设和部署约束为基础, 为在边缘云环境中选择合适的策略提供了原则基础, 并激发了更强大、更通用的训练设计。

3.1 高效端云协同的剪枝和量化策略

在端云协同训练框架中, 剪枝和量化是构建轻量级模型的关键技术, 可以协调边缘设备的有限资源与云模型的准确性要求。李等人。[120]引入了一种稀疏感知通道修剪方法, 该方法评估特征分布偏差以删除全局不重要的通道; 其软掩模机制允许选择性重新激活。EfficientLLM [121] 在预训练期间应用渐进式结构修剪来消除冗余子结构。补充修剪, Jiang等人。[122]提出了一种分割变压器设计, 其中轻量级边缘编码器生成用于云解码的量化中间嵌入。MergeNet [67] 通过 LLM 和 SLM 参数的低秩分解解决结构不兼容问题, 然后进行基于注意力的融合, 从而在不改变边缘架构的情况下实现表达性知识转移。如图 8 所示, 这些方法在训练、推理和调度维度上共同优化剪枝和量化。对于任务粒度混合物

edge-cloud inference, the OT-GAH algorithm [118] leverages online tree pruning and generalized assignment heuristics to achieve near-optimal throughput under diverse latency and accuracy constraints.

3.2 Distillation and Low-Rank Approximation

Edge-cloud distillation techniques fall into three broad categories: 1) task- and domain-adaptive distillation, 2) architectural decoupling with proxy learning, and 3) bidirectional, feedback-driven distillation.

Task- and Domain-Adaptive Distillation. ATKD redefines distillation along task- and diversity-oriented axes [112], introducing an uncertainty coefficient to quantify token-level learning difficulty and revealing that high LLM output certainty suppresses diversity. Methods such as SLMREC [91] and GKT [16] align intermediate hidden states between lightweight on-device SLMs and cloud LLMs for latent-space transfer. Domain-robust techniques [76, 81] construct a pseudo-sample space, using latent mapping, masking, and Mixup, to align student representations without shared data. DDK [101] further improves cross-domain robustness via domain-guided sampling and factor-smoothing mechanism to facilitate efficient cloud-to-edge distillation across diverse domains.

Architectural Decoupling and Feedback-Guided Distillation. To decouple deployment between edge and cloud, DC-CCL [9] vertically partitions a base model into cloud and edge components, training a lightweight proxy via distillation to mimic cloud behavior with minimal communication. Starodubcev *et al.* [108] adaptively invoke teacher corrections only when student outputs fail quality checks. Collin *et al.* [93] integrate weak supervision and high-capacity teachers into a closed-loop framework, while Co-Supervised Learning [102] dynamically allocates teachers and filters noisy signals via posterior consistency. Inverted supervision methods like SALT [107] and SKD [212] have edge SLMs teach LLMs early in training, gradually reversing roles to focus LLMs on complex samples. Multi-modal, feedback-enhanced pipelines, CD-CCA [48] and LLM-QAT [103], upload utility-selected samples for cloud optimization and return quantized updates to the edge, enabling efficient, compressible, and adaptive model enhancement. Collectively, these methods demonstrate a progression from static partitioning to dynamic, bidirectional, and multi-modal knowledge transfer in edge-cloud collaborative training.

Leveraging Low-Rank Approximation for Edge-side SLMs Deployment. Beyond knowledge distillation, low-rank approximation has emerged as a lightweight and effective approach for optimizing LLM deployment in resource-constrained edge environments. QLLMS [294] addresses edge performance unpredictability by reconstructing the AQS matrix from partial samples via a low-rank attribute-driven recovery method. DP-LoRA [293] reduces transmission overhead in distributed training through low-rank weight updates. Moreover, [295] enhances stability-aware fine-tuning using fractional programming and an iterative-level penalty (IRP) to jointly optimize resource allocation and user-edge association.

3.3 Parameter Compatibility and Model Convergence

Traditional unified model-transfer methods struggle with parameter mismatches and personalized edge requirements. To address fragmented domain knowledge in collaborative learning, recent studies propose efficient parameter-coordination mechanisms: Graft [277] introduces a compatibility-aware stitching strategy, using local functional attribution and global information-theoretic signals to integrate only compatible parameters. CKI [68] evaluates a source model’s information capacity and fuses compatible parameters into the target via a two-stage transfer and stitching process. Forward-OFA [69] uses real-time edge demands to assemble sub-networks through behavior-to-structure mapping, resolving gradient conflicts and allowing one-pass adaptation without back-propagation. DIET [70] maintains a unified backbone across devices, while the cloud generates personalized “diets” (subnetworks) based on each device’s usage history. FedMKT [46] explores two-way knowledge exchange between cloud LLMs and client SLMs, improving adaptation under non-IID data. ModelGPT [74] leverages LLMs to automatically generate customized small models from user-provided samples or task descriptions.

边云推理中，OT-GAH 算法 [118] 利用在线树修剪和广义分配启发式方法，在不同的延迟和准确性约束下实现接近最优的吞吐量。

3.2 蒸馏和低阶近似

边缘云蒸馏技术分为三大类：1) 任务和领域自适应蒸馏，2) 与代理学习的架构解耦，以及3) 双向、反馈驱动的蒸馏。

任务和领域自适应蒸馏。ATKD 沿着任务和多样性导向的轴重新定义了蒸馏[112]，引入不确定性系数来量化令牌级别的学习难度，并揭示了 LLM 输出的高确定性抑制了多样性。SLMREC [91] 和 GKT [16] 等方法对齐轻量级设备上 SLM 和云 LLM 之间的中间隐藏状态，以实现潜在空间传输。领域鲁棒技术 [76, 81] 使用潜在映射、掩蔽和混合来构造伪样本空间，以在没有共享数据的情况下对齐学生表示。DDK [101]通过域引导采样和因子平滑机制进一步提高跨域鲁棒性，以促进跨不同域的高效从云到边缘的蒸馏。

架构解耦和反馈引导的蒸馏。为了解耦边缘和云之间的部署，DC-CCL [9] 将基础模型垂直划分为云和边缘组件，通过蒸馏训练轻量级代理，以最少的通信来模拟云行为。斯塔罗杜布切夫等人。[108]仅当学生输出未通过质量检查时才自适应地调用教师纠正。科林等人。[93]将弱监督和高能力教师整合到闭环框架中，而共同监督学习[102]通过后验一致性动态分配教师并过滤噪声信号。像 SALT [107] 和 SKD [212] 这样的反向监督方法让边缘 SLM 在训练早期教导 LLM，逐渐反转角色，让 LLM 专注于复杂样本。多模式、反馈增强的管道、CD-CCA [48] 和 LLM-QAT [103]，上传实用程序选择的样本以进行云优化，并将量化更新返回到边缘，从而实现高效、可压缩和自适应模型增强。总的来说，这些方法展示了边缘云协作训练中从静态分区到动态、双向和多模式知识转移的进展。

利用低秩近似进行边缘侧 SLM 部署。。除了知识蒸馏之外，低秩近似已成为一种轻量级且有效的方法，用于在资源受限的边缘环境中优化 LLM 部署。QLLMS [294] 通过低秩属性驱动的恢复方法从部分样本重建 AQS 矩阵，解决了边缘性能的不可预测性。DP-LoRA [293]通过低秩权重更新减少了分布式训练中的传输开销。此外，[295]使用分数规划和迭代级惩罚（IRP）增强了稳定性感知微调，以共同优化资源分配和用户边缘关联。

3.3 参数兼容性和模型收敛性

传统的统一模型传输方法面临着参数不匹配和个性化边缘要求的问题。为了解决协作学习中的碎片化领域知识，最近的研究提出了有效的参数协调机制：Graft [277]引入了一种兼容性感知缝合策略，使用局部功能归因和全局信息论信号来仅集成兼容的参数。CKI [68]评估源模型的信息容量，并通过两阶段传输和拼接过程将兼容参数融合到目标中。Forward-OFA [69]使用实时边缘需求通过行为到结构映射来组装子网络，解决梯度冲突并允许无需反向传播的一次性适应。DIET [70] 维护跨设备的统一主干，而云根据每个设备的使用历史生成个性化的“饮食”（子网）。FedMKT [46] 探索了云 LLM 和客户端 SLM 之间的双向知识交换，提高了非 IID 数据下的适应性。ModelGPT [74] 利用 LLM 从用户提供的样本或任务描述自动生成定制的小模型。

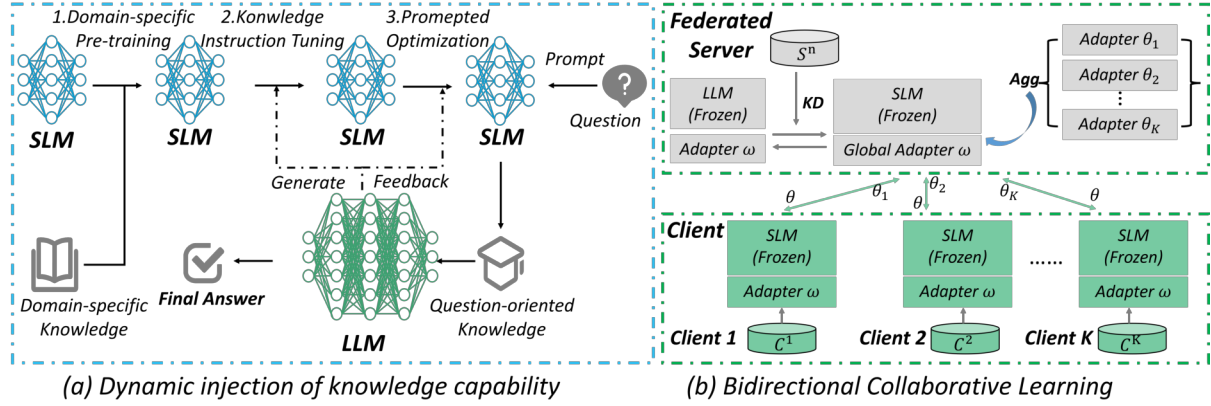


Fig. 9. Collaborative training with bidirectional knowledge transfer: (a) the SLM acquires task knowledge through domain-specific pretraining, instruction tuning, and prompt optimization; (b) frozen client SLMs augmented with local adapters train collaboratively under centralized coordination.

3.4 Adapter-Based Modular Designs

In collaborative edge-cloud systems, integrating knowledge and weight parameters between SLMs and LLMs remains a core challenge. To address this, adapter methods (*e.g.*, LoRA) enable scalable, parameter-efficient tuning across heterogeneous edge-cloud systems: PEFT [79] uses lightweight LoRA modules as communication bridges: clients update only adapter weights during federated training, while the server performs bidirectional knowledge extraction via supervised fine-tuning and KL-divergence regularization. Lu *et al.* [280] insert bottleneck adapters between Transformer layers to encode domain knowledge into pre-trained models. PLURALISM [88] integrates community LLMs into a foundation model via LoRA, enabling edge SLMs to fine-tune locally and the cloud LLM to manage integration. HETLoRA [96] combines high- and low-rank LoRA modules; clients rank modules by capability, and the server applies rank-aware pruning and sparsity-weighted aggregation. CDC-MMPG [36] introduces a Fast Domain Adapter that uses historical multi-modal data to train a global model and generates personalized parameters for real-time device inputs. Collectively, these approaches mark a shift toward flexible, modular, and data-aware tuning mechanisms that seamlessly integrate knowledge across end-cloud collaborative frameworks.

3.5 Bidirectional Collaborative Learning with Heterogeneous Models

Most distribution-shift methods rely on on-device personalization via online training or gradient updates, which incur high latency and overhead. To address this, recent approaches favors collaborative optimization across heterogeneous edge-cloud models, emphasizing iterative, bidirectional knowledge transfer over static, one-way distillation (see Fig. 9): DUET [104] separates static and dynamic layers, using a Universal Meta-Network and hierarchical HyperNetworks to generate adaptive parameters for personalized edge inference. Hu *et al.* [270] develop a knowledge-transfer prompting technique that lets different LLMs share and integrate complementary knowledge effectively. CROSSLM [87] establishes a pipeline where edge models train locally, and the cloud generates and filters pseudo-data using edge feedback, enabling mutual enhancement. SLM [89] applies inverse reinforcement learning on the edge to produce structured, task-aligned samples, while cloud LLMs perform parallel-decoding distillation to align global and local objectives.

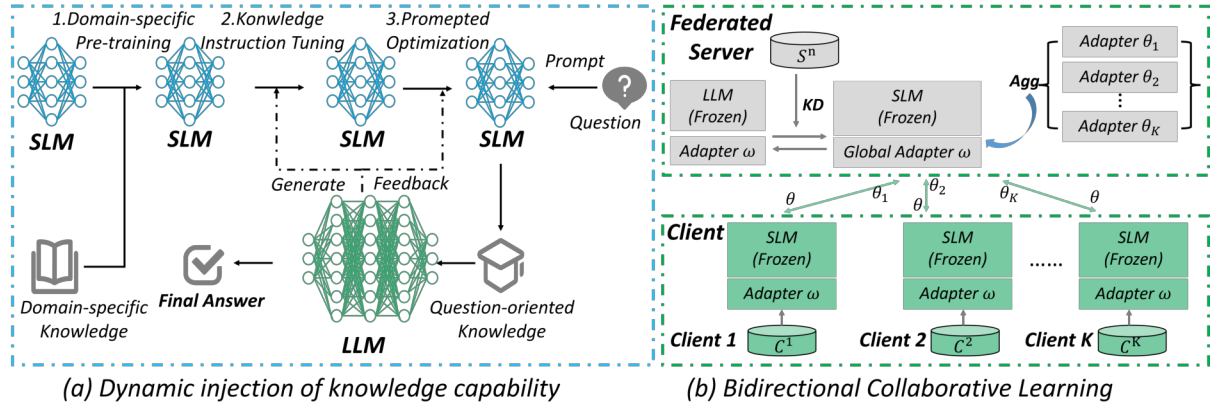


图 9. 双向知识迁移的协作训练: (a) SLM 通过特定领域的预训练、指令调整和提示优化来获取任务知识; (b) 冻结的客户端 SLM 增强了本地适配器, 在集中协调下进行协作训练。

3.4 基于适配器的模块化设计

在协作边缘云系统中, 集成 SLM 和 LLM 之间的知识和权重参数仍然是一个核心挑战。为了解决这个问题, 适配器方法 (例如 LoRA) 可以跨异构边缘云系统进行可扩展、参数高效的调整: PEFT [79] 使用轻量级 LoRA 模块作为通信桥梁: 客户端在联合训练期间仅更新适配器权重, 而服务器通过监督微调 and K L 散度正则化执行双向知识提取。卢等人。[280] 在 Transformer 层之间插入瓶颈适配器, 将领域知识编码到预先训练的模型中。PLURALISM [88] 通过 LoRA 将社区 LLM 集成到基础模型中, 使边缘 SLM 能够在本地进行微调, 云 LLM 能够管理集成。HETLoRA [96] 结合了高阶和低阶 LoRA 模块; 客户端按功能对模块进行排名, 服务器应用排名感知修剪和稀疏加权聚合。CDC-MMPG [36] 引入了一种快速域适配器, 它使用历史多模态数据来训练全局模型并为实时设备输入生成个性化参数。总的来说, 这些方法标志着向灵活、模块化和数据感知调整机制的转变, 这些机制可以跨端云协作框架无缝集成知识。

3.5 异构模型的双向协作学习

大多数分布转移方法依赖于通过在线训练或梯度更新进行的设备上个性化, 这会产生高延迟和开销。为了解决这个问题, 最近的方法有利于跨异构边缘云模型的协作优化, 强调迭代、双向知识转移而不是静态、单向蒸馏 (见图 9): DUET [104] 分离静态层和动态层, 使用通用元网络和分层超网络来生成用于个性化边缘推理的自适应参数。胡等人。[270] 开发一种知识转移促进技术, 让不同的大语言模型有效地共享和整合互补知识。CROSSLM [87] 建立了一个管道, 其中边缘模型在本地进行训练, 云使用边缘反馈生成和过滤伪数据, 从而实现相互增强。SLM [89] 在边缘应用逆强化学习来生成结构化的、与任务一致的样本, 而云 LLM 则执行并行解码蒸馏以协调全局和局部目标。

3.5.1 Dynamic Knowledge Transfer from Cloud LLMs to On-Device SLMs. This research line treats small models as pluggable knowledge modules that supplement or enhance LLM performance during inference. Knowledge Card [176] uses a pretrained SLM to generate “knowledge cards”, a parameterized knowledge base dynamically queried during inference. BLADE [181] employs a plugin-style architecture where domain-specific SLMs capture expert knowledge and fuse it with general-purpose LLMs via Bayesian optimization to improve reasoning quality. Progressive Distillation [99] targets unlabeled data by having cloud LLMs generate task labels and rationales to train lightweight edge models, jointly optimizing label and rationale losses for improved accuracy and interpretability. The HEF framework [180] integrates a small-scale Empathy Model to detect emotions and triggers, guiding LLM generation for more nuanced dialogue. FedCFA [73] addresses non-IID generalization in federated settings by constructing a global feature dataset and applying causality-aware counterfactual replacement, aligning SLMs with cloud semantics. AcKnowledge [281] leverages user-question-driven meta-learning to acquire external knowledge and continually refine SLMs with user feedback when distortions are detected.

3.5.2 Adaptive Capability Injection into Small Models via Cloud Guidance. This category reverses the typical distillation flow, using SLMs as “teachers” or behavioral priors to guide LLM training and fine-tuning. ECLM [256] decomposes cloud models into composable modules, assembling task-specific submodels for diverse devices and periodically integrating new edge knowledge back into the cloud. Mitchell *et al.* [105] encode fine-tuning-induced behavioral changes as a compact “logit delta”, which a lightweight SLM computes and transfers to an LLM to emulate these adaptations without full retraining. Tang *et al.* [179, 206] present an entity-relation extraction system where a small PLM captures head-class knowledge and generates intermediate predictions serving as chain-of-thought guidance for the LLM. Purify-LLM [177] filters noisy data for LLMs using a trusted SLM and the CP- δ algorithm, integrating only aligned knowledge. These methods highlight SLMs’ role in constraining and guiding LLM learning, creating a positive feedback loop that optimizes model behavior along explicit paths.

4 Benchmarks, Datasets, and Evaluation Protocols

Evaluating collaborative LLM-SLM systems is more complex than centralized models due to a lack of standardized benchmarks for edge-cloud collaboration. Traditional NLP benchmarks assume IID data and centralized execution, unlike real deployments where SLMs manage personalized, non-IID workloads. Thus, new benchmarks must enable user- or device-level partitioning to reflect deployment heterogeneity.

4.1 Datasets for Federated Edge-Cloud Collaboration

Realistic, heterogeneous, and reproducible benchmarks are crucial for evaluating collaborative training and inference in edge-cloud LLM-SLM systems (see Table 7). The LEAF framework [145] provides modular tools and real-world, user-partitioned datasets to assess performance under statistical heterogeneity, resource constraints, and personalization. For large-scale visual tasks, Zhu *et al.* [146] introduce iNaturalist-User-120k (120K images, 1.2K species, 9.3K users) and Landmarks-User-160k (164K images, 2K categories, 1.3K uploaders) with non-IID partitions. In dialogue modeling, PersonalDialog [147] offers 56 M utterances from 8.5 M speakers annotated with demographic attributes, while PERSONA-CHAT [148] provides persona-based consistency benchmarks. LiveChat [149] supplies 1.33 M multi-party Chinese dialogues with 351 roles, enabling response generation and recipient identification under temporal dynamics. FedScale [150] covers 18 real-world tasks across modalities and scales from hundreds to millions of clients, with its FAR platform supporting asynchronous training, latency simulation, and metrics for accuracy, cost, and privacy. FedNLP [151] benchmarks federated NLP tasks (*e.g.*, 20Newsgroup, MRQA) using Dirichlet splits to simulate non-IID distributions, facilitating reproducible evaluation and real-world optimization of collaborative LLM-SLM systems.

3.5.1 从云 LLM 到设备上 SLM 的动态知识传输。该研究线将小型模型视为可插入的知识模块，以补充或增强推理过程中的 LLM 性能。知识卡[176]使用预训练的SLM来生成“知识卡”，这是在推理过程中动态查询的参数化知识库。BLADE [181] 采用插件式架构，其中特定领域的 SLM 捕获专家知识，并通过贝叶斯优化将其与通用 LLM 融合，以提高推理质量。渐进蒸馏[99]通过让云大语言模型生成任务标签和基本原理来训练轻量级边缘模型，联合优化标签和基本原理损失以提高准确性和可解释性，从而针对未标记数据。HEF 框架 [180] 集成了一个小型移情模型来检测情绪和触发因素，指导 LLM 生成进行更细致的对话。FedCFA [73] 通过构建全局特征数据集并应用因果感知反事实替换，使 SLM 与云语义保持一致，解决联邦设置中的非独立同分布泛化问题。AcKnowledge [281]利用用户问题驱动的元学习来获取外部知识，并在检测到扭曲时根据用户反馈不断完善SLM。

3.5.2 通过云指导将自适应能力注入小型模型。这一类别反转了典型的蒸馏流程，使用 SLM 作为“老师”或行为先验来指导 LLM 培训和微调。ECLM [256]将云模型分解为可组合模块，为不同设备组装特定于任务的子模型，并定期将新的边缘知识集成回云中。米切尔等人。[105] 将微调引起的行为变化编码为紧凑的“logit delta”，轻量级 SLM 对其进行计算并传输到 LLM 以模拟这些适应，而无需完全重新训练。唐等人。[179, 206] 提出了一种实体关系提取系统，其中小型 PLM 捕获头级知识并生成中间预测，作为 LLM 的思想链指导。Purify-LLM [177] 使用可信的 SLM 和 CP- δ 算法过滤 LLM 的噪声数据，仅集成对齐的知识。这些方法强调了 SLM 在约束和指导 LLM 学习方面的作用，创建了一个沿着显式路径优化模型行为的正反馈循环。

4 基准、数据集和评估协议

由于缺乏边缘-云协作的标准化基准，评估协作式 LLM-SLM 系统比集中式模型更复杂。传统的 NLP 基准假设 IID 数据和集中执行，这与 SLM 管理个性化、非 IID 工作负载的实际部署不同。因此，新的基准必须支持用户或设备级分区以反映部署异构性。

4.1 边云联合协作数据集

现实、异构且可重复的基准对于评估边缘云 LLM-SLM 系统中的协作训练和推理至关重要（参见表 7）。LEAF 框架 [145] 提供模块化工具和真实世界的用户分区数据集，以评估统计异质性、资源限制和个性化下的性能。对于大规模视觉任务，Zhu 等人。[146]引入非IID分区的iNaturalist-User-120k（120K图像，1.2K物种，9.3K用户）和Landmarks-User-160k（164K图像，2K类别，1.3K上传者）。在对话建模中，PersonalDialog [147] 提供来自 850 万说话者的 5600 万条话语，并带有人口统计属性注释，而 PERSON A-CHAT [148] 提供基于角色的一致性基准。LiveChat [149] 提供 133 万多方中文对话和 351 个角色，能够在时间动态下实现响应生成和接收者识别。FedScale [150] 涵盖了跨模式的 18 项现实世界任务，规模从数百到数百万客户，其 FAR 平台支持异步训练、延迟模拟以及准确性、成本和隐私指标。FedNLP [151] 使用 Dirichlet 分割来模拟非 IID 分布，对联合 NLP 任务（例如 20Newsgroup、MRQA）进行基准测试，促进协作 LLM-SLM 系统的可重复评估和实际优化。

Table 7. Representative datasets for edge-cloud collaboration

Dataset	Task	Volume & Classes	Partitioning	Metrics & Features
LEAF [145]	Multi-task federated learning	Sent140: 660k tweets; FEMNIST: 805k images	User-level splits reflecting real-world heterogeneity	Supports personalization and transfer learning
iNaturalist-User-120k [146]	Image classification	120,300 images from 9,275 users; 1,203 species	Uploader-ID splits (user-level)	Large label space with strong user-level statistical skew
Landmarks-User-160k [146]	Landmark recognition	164,172 images from 1,262 photographers; 2,028 landmarks	Region-aware S2-cell splits via GPS metadata	Captures spatial distribution heterogeneity
PersonalDialog [147]	Personalized dialogue	56.3 M utterances from 8.5 M speakers	User-ID splits with demographic annotations	Supports sociolinguistic and attribute personalization
PERSONA-CHAT [148]	Persona-grounded chit-chat	164,356 utterances across 10,981 dialogues	Crowdworker pairs on 1,155 personas	Next-utterance prediction and personal consistency
LiveChat [149]	Multi-party live dialogue	1.33 M utterances from 351 streamers	ASR-aligned utterances with recipient matching	Single-turn dialogue with personas; addressee recognition
FedScale [150]	Multi-domain federated learning	20 datasets (CV, NLP, audio) with millions of clients	Non-IID splits by user ID or trace logs	Benchmarks statistical; System heterogeneous, dynamics
FedNLP [151]	Federated NLP (TC, NER, QA, Seq2Seq)	20News: 11.3k, 20 classes; OntoNotes: 50k, 37 tags	Dirichlet non-IID splits and cross-silo partitions	Metrics: accuracy, F1, ROUGE; federated evaluation

4.2 Benchmarks for Dual-Model Edge-Cloud Learning

Edge-cloud collaborative learning demands dual evaluation: global generalization by the cloud LLM and local adaptation by edge SLMs, under non-IID data across devices. Recent benchmarks create non-IID test environments via user-level splits. LEAF extends CIFAR-10 and Shakespeare into federated versions with device-ID partitions [145]. Persona-Chat [148] and LiveChat [149] exploit dialogue roles for personalized generation. FedMulti-modal [152] integrates ten datasets across eight modalities, simulating missing modalities and label noise. FederatedScope-GNN [160] offers heterogeneous graph benchmarks (*e.g.*, FedDBLP) with node-, edge-, and graph-level splits. pFL-Bench [153] unifies partition strategies across 10+ datasets and introduces device heterogeneity and sparsity configurations for realism.

Hierarchical evaluation metrics are vital: image classification employs weighted global accuracy (by device data volume), local SLM accuracy, and distillation loss to assess knowledge transfer [158]. For NLG, global BLEU/ROUGE scores complement user-level perplexity. In recommendations, global AUC weighted by user data and local CTR correction reflect personalization [154], with OCPC evaluating end-to-end traffic-allocation efficiency [155].

表 7. 边云协作的代表性数据集

Dataset	Task	Volume & Classes	Partitioning	Metrics & Features
LEAF [145]	Multi-task federated learning	Sent140: 660k tweets; FEMNIST: 805k images	User-level splits reflecting real-world heterogeneity	Supports personalization and transfer learning
iNaturalist-User-120k [146]	Image classification	120,300 images from 9,275 users; 1,203 species	Uploader-ID splits (user-level)	Large label space with strong user-level statistical skew
Landmarks-User-160k [146]	Landmark recognition	164,172 images from 1,262 photographers; 2,028 landmarks	Region-aware S2-cell splits via GPS metadata	Captures spatial distribution heterogeneity
PersonalDialog [147]	Personalized dialogue	56.3 M utterances from 8.5 M speakers	User-ID splits with demographic annotations	Supports sociolinguistic and attribute personalization
PERSONA-CHAT [148]	Persona-grounded chit-chat	164,356 utterances across 10,981 dialogues	Crowdworker pairs on 1,155 personas	Next-utterance prediction and personal consistency
LiveChat [149]	Multi-party live dialogue	1.33 M utterances from 351 streamers	ASR-aligned utterances with recipient matching	Single-turn dialogue with personas; addressee recognition
FedScale [150]	Multi-domain federated learning	20 datasets (CV, NLP, audio) with millions of clients	Non-IID splits by user ID or trace logs	Benchmarks statistical; System heterogeneous, dynamics
FedNLP [151]	Federated NLP (TC, NER, QA, Seq2Seq)	20News: 11.3k, 20 classes; OntoNotes: 50k, 37 tags	Dirichlet non-IID splits and cross-silo partitions	Metrics: accuracy, F1, ROUGE; federated evaluation

4.2 双模型边云学习基准

边缘-云协作学习需要双重评估：在跨设备的非 IID 数据下，云 LLM 的全局泛化和边缘 SLM 的本地适应。最近的基准测试通过用户级拆分创建非独立同分布测试环境。LEAF 将 CIFAR-10 和 Shakespeare 扩展为具有设备 ID 分区的联合版本 [145]。Persona-Chat [148] 和 LiveChat [149] 利用对话角色进行个性化生成。FedMulti-modal [152] 集成了八种模态的十个数据集，模拟缺失的模态和标签噪声。FederatedScopeGNN [160] 提供具有节点、边缘和图级别拆分的异构图基准（例如 FedDBLP）。pFL-Bench [153] 统一了 10+ 数据集的分区策略，并引入了设备异构性和稀疏性配置以实现真实性。

分层评估指标至关重要：图像分类采用加权全局精度（按设备数据量）、局部 SLM 精度和蒸馏损失来评估知识转移 [158]。对于 NLG，全局 BLEU/ROUGE 分数补充了用户级别的困惑。在推荐中，由用户数据加权的全局 AUC 和本地 CTR 校正反映了个性化[154]，OCPC 评估端到端流量分配效率[155]。

Open-source platforms support reproducible edge-cloud experiments: FedML [159] offers on-device, single-machine, and distributed modes with modular communication-training decoupling; Flower [161] scales to millions of virtual clients across languages and backends; Google’s TensorFlow Federated [158] runs on hundreds of millions of devices. For LLM-specific operations, LLMOps frameworks [156, 164] introduce tailored monitoring and safety protocols. SpecBench [244] evaluates speculative decoding across latency, accuracy, and compute cost. MessageRewriteVal [227] standardizes mobile text rewriting with high-quality, human-annotated scenarios to assess a model’s ability to rewrite messages based on natural language instructions. Together, these benchmarks and platforms establish a foundation for fair, rigorous evaluation of collaborative LLM-SLM systems under realistic edge-cloud conditions.

5 Open Challenges

5.1 Privacy-Preserving and Secure Collaboration Mechanisms

Traditional optimizations for inference efficiency often overlook privacy and security risks in collaborative training and updating [113]. Achieving efficient edge-cloud cooperation while preserving data locality has thus become a critical research direction, with secure multi-party computation encrypting gradient updates to prevent sensitive information leakage during aggregation.

Training-level protections. Liu *et al.* [57] integrate edge-side prompt adaptation with cloud-based optimization via lightweight adapters, enabling privacy-preserving collaboration by keeping sensitive data local while leveraging cloud resources for efficiency. DCPR [63] adopts a hierarchical diffusion paradigm for personalized recommendation, where general patterns are learned in the cloud, regional preferences are adapted at the edge, and fine-grained personalization occurs on-device. Luo *et al.* [278] utilize federated learning to train models across decentralized clients, ensuring privacy by transmitting only encrypted updates. Building upon these advances, future efforts will increasingly center on uncertainty-guided supervision for selective cloud involvement, fair aggregation under skewed and non-*i.i.d.* client distributions, and generalizable knowledge transfer across heterogeneous tasks, modalities, and model scales. Furthermore, counterfactual representation learning is expected to play a pivotal role in debiasing cloud-side updates before aggregation, enhancing both utility and fairness in collaborative training.

Inference-level protections. RemoteRAG [247] introduces a semantic differential-privacy metric in embedding space and a dynamic secure retrieval protocol that only returns document indices when similarity exceeds a threshold. SuperICL [58] uses plugin-style small models to inject local knowledge via prompt augmentation, guiding the LLM’s reasoning without exposing raw data. Pan *et al.* [35] propose a hybrid framework where the cloud builds proxy models with adapters and compression, and the edge applies symbolic masking to sensitive content, generating substitute data via cGANs to balance utility and privacy. POST [222] accelerates privacy-preserving speculative decoding by offloading draft generation to a public GPT model and optimizing cryptographic operations. Chen *et al.* [197] orchestrate a hybrid Kubernetes-based system that retains sensitive data and vector stores (*e.g.*, Chroma) in private clouds while elastically offloading LLM inference to public clouds under strict confidentiality constraints. These efforts provide initial approaches to privacy-preserving collaboration but reveal open challenges. Future work should scale secure, efficient speculative decoding across diverse model hierarchies, balance draft quality and verification cost in multi-model setups, ensure robustness amid noisy feedback and unstable connections, and enable emergent capabilities through self-reflective SLMs within larger collaborative systems.

开源平台支持可重复的边缘云实验：FedML [159] 提供具有模块化通信训练解耦的设备上、单机和分布式模式；Flower [161] 可跨语言和后端扩展到数百万个虚拟客户端；Google 的 TensorFlow Federated [158] 在数亿台设备上运行。对于 LLM 特定的操作，LLMOps 框架 [156, 164] 引入了定制的监控和安全协议。SpecBench [244] 评估推测解码的延迟、准确性和计算成本。MessageRewriteVal [227] 使用高质量、人工注释的场景标准化移动文本重写，以评估模型基于自然语言指令重写消息的能力。这些基准和平台共同为在现实边缘云条件下对协作式 LLM-SLM 系统进行公平、严格的评估奠定了基础。

5 项公开挑战

5.1 隐私保护和协作机制

传统的推理效率优化往往忽视协作训练和更新中的隐私和安全风险[113]。因此，在保持数据局部性的同时实现高效的边云合作已成为一个关键的研究方向，安全多方计算加密梯度更新以防止聚合过程中敏感信息泄漏。

培训级保护。刘等人。[57]通过轻量级适配器将边缘端即时适应与基于云的优化集成起来，通过将敏感数据保留在本地来实现隐私保护协作，同时利用云资源提高效率。DCPR [63] 采用分层扩散范式进行个性化推荐，其中一般模式在云端学习，区域偏好在边缘进行调整，细粒度个性化在设备上。罗等人。[278]利用联合学习来训练分散客户端的模型，通过仅传输加密更新来确保隐私。在这些进步的基础上，未来的工作将越来越集中于不确定性引导的监督，以选择性云参与、倾斜和非独立同分布下的公平聚合。客户分布，以及跨异构任务、模式和模型规模的通用知识转移。此外，反事实表示学习预计将在聚合前消除云端更新偏差方面发挥关键作用，从而增强协作训练的实用性和公平性。

推理级保护。RemoteRAG [247] 在嵌入空间中引入了语义差异隐私度量和动态安全检索协议，该协议仅在相似性超过阈值时返回文档索引。SuperICL [58]使用插件式的小模型通过即时增强注入局部知识，在不暴露原始数据的情况下指导大语言模型的推理。潘等人。[35]提出了一种混合框架，其中云通过适配器和压缩构建代理模型，边缘对敏感内容应用符号屏蔽，通过 cGAN 生成替代数据以平衡实用性和隐私。POST [222] 通过将草案生成卸载到公共 GPT 模型并优化加密操作来加速隐私保护推测解码。陈等人。[197]编排一个基于 Kubernetes 的混合系统，该系统在私有云中保留敏感数据和向量存储（例如 Chroma），同时在严格的保密约束下将 LLM 推理弹性卸载到公共云。这些努力为隐私保护协作提供了初步方法，但也揭示了开放的挑战。未来的工作应该在不同的模型层次结构中扩展安全、高效的推测性解码，平衡多模型设置中的草稿质量和验证成本，确保在嘈杂的反馈和不稳定的连接中的鲁棒性，并通过更大的协作系统中的自反射 SLM 实现紧急功能。

Table 8. Industrial cloud-edge LLM-SLM collaboration frameworks and application scenarios

Category	Domain	Collaboration Strategy	Advantages & Limitations
Industrial Edge-Cloud Frameworks	Walle [264]	End-to-end deployment pipeline handling development, runtime, and scaling.	Manages 300+ tasks and 10B+ daily calls; constrained by tight infrastructure coupling.
	Luoxi [265]	Slow-fast learning: cloud LLM generates aux. reps; edge SLM fast inference with feedback.	Enhances personalization via feedback loop; requires robust bidirectional communication.
	InfGUIAgent [266]	Two-stage hierarchical reasoning with on-device fine-tuning for multi-modal GUI interaction.	Enables on-device GUI reasoning; may face edge memory and compute limitations.
Vertical Applications	Autonomous driving [55, 56]	Edge SLM handles perception; cloud LLM performs high-level planning and reasoning.	Balances fast perception with complex strategy; requires efficient cloud triggering.
	Livestream product recognition [184]	Edge extracts keyframe features; cloud LLM conducts multi-modal classification.	Reduces bandwidth via selective uploads; risks missing critical content in keyframes.
	Cultural heritage restoration [77]	Edge proposes fragment matches; cloud LLM ranks and aligns results.	Supports time-sensitive, high-accuracy restoration; depends on fragment proposal quality.
	Product modeling[78]	Edge injects geometric priors; cloud completes texture synthesis.	Improves partial-view generation; limited by cross-device alignment noise.
	Virtual assistants [240, 241]	Edge SLM manages routine interactions; cloud LLM handles complex or fallback queries.	Ensures responsiveness; cloud fallback may disrupt conversational continuity.
	Personalized recommendation [30, 63, 221]	Cloud LLM generates candidate items; edge re-ranks using real-time user context.	Adapts swiftly to user intent; candidate diversity hinges on LLM quality.
	Mobile task automation [224]	Cloud LLM parses app semantics; edge executes tasks via dynamic analysis.	Enables cross-app automation without manual setup; depends on accurate semantic parsing.
	Healthcare [225]	Biomedical LLM fine-tuned from general LLM for on-premise or hybrid deployment.	Offers domain-specific open-source model; needs careful tuning for sensitive settings.
	Web interaction [230]	Cloud LLM decomposes tasks and generates executable scripts.	Automates multi-step web tasks; relies on precise decomposition.

表8. 工业云边缘LLM-SLM协作框架及应用场景

Category	Domain	Collaboration Strategy	Advantages & Limitations
d u o C e g d E i a r r i s u d n i s k r o w e m a r f	Walle [264]	End-to-end deployment pipeline handling development, runtime, and scaling.	Manages 300+ tasks and 10B+ daily calls; constrained by tight infrastructure coupling.
	Luoxi [265]	Slow-fast learning: cloud LLM generates aux. reps; edge SLM fast inference with feedback.	Enhances personalization via feedback loop; requires robust bidirectional communication.
	InfGUIAgent [266]	Two-stage hierarchical reasoning with on-device fine-tuning for multi-modal GUI interaction.	Enables on-device GUI reasoning; may face edge memory and compute limitations.
s n o i a c i p A i a c t i r e V	Autonomous driving [55, 56]	Edge SLM handles perception; cloud LLM performs high-level planning and reasoning.	Balances fast perception with complex strategy; requires efficient cloud triggering.
	Livestream product recognition [184]	Edge extracts keyframe features; cloud LLM conducts multi-modal classification.	Reduces bandwidth via selective uploads; risks missing critical content in keyframes.
	Cultural heritage restoration [77]	Edge proposes fragment matches; cloud LLM ranks and aligns results.	Supports time-sensitive, high-accuracy restoration; depends on fragment proposal quality.
	Product modeling[78]	Edge injects geometric priors; cloud completes texture synthesis.	Improves partial-view generation; limited by cross-device alignment noise.
	Virtual assistants [240, 241]	Edge SLM manages routine interactions; cloud LLM handles complex or fallback queries.	Ensures responsiveness; cloud fallback may disrupt conversational continuity.
	Personalized recommendation [30, 63, 221]	Cloud LLM generates candidate items; edge re-ranks using real-time user context.	Adapts swiftly to user intent; candidate diversity hinges on LLM quality.
	Mobile task automation [224]	Cloud LLM parses app semantics; edge executes tasks via dynamic analysis.	Enables cross-app automation without manual setup; depends on accurate semantic parsing.
	Healthcare [225]	Biomedical LLM fine-tuned from general LLM for on-premise or hybrid deployment.	Offers domain-specific open-source model; needs careful tuning for sensitive settings.
	Web interaction [230]	Cloud LLM decomposes tasks and generates executable scripts.	Automates multi-step web tasks; relies on precise decomposition.

5.2 Application of Edge-Cloud Large-Small Models in Vertical Domains

Edge-cloud LLM-SLM collaborations demonstrate strong potential across diverse application domains (see Table 8). In autonomous driving, ADAS [55] uses a lightweight multi-modal model (CogVLM2) at the edge for latency-critical tasks (object detection, obstacle avoidance) while a cloud LLM (ChatGPT-4o) handles high-level route planning. EC-Drive [56] extends this with an event-driven design: the edge LLM manages routine driving, triggering the cloud LLM upon distribution shifts (*e.g.*, novel obstacles) for commonsense reasoning and cross-modal understanding.

Live streaming challenges real-time product recognition with low latency, high concurrency, and costly multi-modal inference. Recent frameworks tackle this by using uni-modal edge models to selectively upload keyframes for cloud multi-modal processing [184]. For example, LLMCO4MR [77] treats manuscript restoration as a combinatorial task, using on-device neural solvers for fragment selection and cloud LLMs for confidence scoring and alignment. BoFiCap [286] decouples image captioning into boundary detection on-device and non-autoregressive filling in the cloud. VITA-1.5 [285] employs a three-stage training pipeline (vision-language pretraining, audio alignment, speech decoding) to avoid modality conflicts, while MPOD123 [78] injects geometric priors at the edge and offloads complex visual generation to the cloud.

Recommendation systems such as *cenarios* [263], LSC4Rec [30] use cloud LLMs to generate diverse candidate lists, with lightweight edge ranking models adapting in real time to user preferences. AutoDroid [224] automates mobile workflows by combining LLM commonsense reasoning with dynamic analysis of app interfaces. In healthcare, BioMistral [225] specializes a general LLM via continued pretraining on biomedical texts to support domain-specific tasks. For web-based interactions, WebAgent [230] addresses open-domain variability and the lack of HTML inductive biases by decomposing user instructions into structured subtasks, summarizing lengthy documents, and generating executable Python scripts to perform complex operations. Together, these vertical applications illustrate the transformative impact of hierarchical edge-cloud LLM-SLM architectures across real-world scenarios.

6 Conclusion

This survey offers the first comprehensive review of edge-cloud collaboration between large and small language models (SLMs), addressing both inference-time cooperation and training-time coordination. We propose a unified taxonomy of inference paradigms, task assignment, task division, and hybrid approaches, further refined into task- and token-level granularities. On the training side, we identify key enablers such as bidirectional knowledge transfer, quantization, pruning, and low-rank adaptation, which support efficient deployment without compromising performance. By grounding our analysis in recent advances, we bridge algorithmic techniques with system-level requirements, providing methodological insights and practical guidance for building scalable, low-latency, resource-aware LLM-SLM systems. This unified perspective establishes a solid foundation for future research and deployment in heterogeneous environments.

Future Prospects. In edge-cloud collaborative inference, many systems rely on uncertainty estimation to support dynamic decisions such as early exit, fallback, or model switching, typically by comparing uncertainty between edge-side SLMs and cloud LLMs. However, mainstream methods based on sampling consistency or token-level probabilities face inherent limitations. Normalized softmax probabilities obscure the raw evidential strength encoded in logits, often failing to distinguish whether a model is confident, uncertain, or unfamiliar with the input. This misalignment is particularly problematic in open-ended language generation, where multiple valid continuations may exist. We advocate a future shift toward evidence-based uncertainty estimation, which leverages unnormalized logits to preserve the model's accumulated training experience. By decomposing uncertainty into epistemic and aleatoric components, *e.g.*, via Dirichlet-based formulations, edge models can more accurately characterize their own reliability. Such evidence-aware strategies offer a finer-grained basis for trust calibration,

5.2 边云大小模型在垂直领域的应用

边缘云 LLM-SLM 协作在不同的应用领域展示了强大的潜力（见表 8）。在自动驾驶中，ADAS [55] 在边缘使用轻量级多模态模型 (CogVLM2) 来执行延迟关键型任务（对象检测、避障），而云 LLM (ChatGPT-4o) 则处理高级路线规划。EC-Drive [56] 通过事件驱动设计扩展了这一点：边缘 LLM 管理日常驾驶，在分布变化（例如，新障碍）时触发云 LLM，以进行常识推理和跨模式理解。

直播对低延迟、高并发和昂贵的多模态推理的实时产品识别提出了挑战。最近的框架通过使用单模态边缘模型来选择性地上传云多模态处理的关键帧来解决这个问题[184]。例如，LLMCO4MR [77] 将手稿恢复视为一项组合任务，使用设备上的神经解算器进行片段选择，并使用云 LLM 进行置信度评分和对齐。BoFiCap [286] 将图像字幕解耦为设备上的边界检测和云中的非自回归填充。VITA-1.5 [285] 采用三阶段训练管道（视觉语言预训练、音频对齐、语音解码）来避免模态冲突，而 MPOD123 [78] 在边缘注入几何先验，并将复杂的视觉生成卸载到云端。

cenarios [263]、LSC4Rec [30] 等推荐系统使用云 LLM 生成不同的候选列表，并使用实时适应用户偏好的轻量级边缘排名模型。AutoDroid [224] 通过将大语言模型常识推理与应用程序界面的动态分析相结合，实现移动工作流程的自动化。在医疗保健领域，BioMistral [225] 通过持续对生物医学文本进行预训练来专门学习一般大语言模型，以支持特定领域的任务。对于基于 Web 的交互，WebAgent [230] 通过将用户指令分解为结构化子任务、总结冗长的文档以及生成可执行的 Python 脚本来执行复杂的操作，解决了开放域可变性和 HTML 归纳偏差的缺乏。这些垂直应用程序共同说明了分层边缘云 LLM-SLM 架构在现实场景中的变革性影响。

6 结论

这项调查首次全面回顾了大型和小型语言模型 (SLM) 之间的边缘云协作，解决了推理时协作和训练时协调问题。我们提出了推理范式、任务分配、任务划分和混合方法的统一分类法，并进一步细化为任务和令牌级别的粒度。在训练方面，我们确定了双向知识转移、量化、修剪和低秩适应等关键推动因素，这些因素支持高效部署而不影响性能。通过基于最新进展的分析，我们将算法技术与系统级需求联系起来，为构建可扩展、低延迟、资源感知的 LLM-SLM 系统提供方法见解和实践指导。这种统一的视角为未来异构环境中的研究和部署奠定了坚实的基础。

前景。在边缘-云协作推理中，许多系统依靠不确定性估计来支持动态决策，例如提前退出、回退或模型切换，通常是通过比较边缘侧 SLM 和云 LLM 之间的不确定性。然而，基于采样一致性或令牌级概率的主流方法面临着固有的局限性。归一化的 softmax 概率掩盖了 logits 中编码的原始证据强度，通常无法区分模型是否对输入有信心、不确定或不熟悉。这种错位在开放式语言生成中尤其成问题，其中可能存在多个有效延续。我们主张未来转向基于证据的不确定性估计，利用非标准化逻辑来保留模型积累的训练经验。通过将不确定性分解为认知和任意成分（例如通过基于狄利克雷的公式），边缘模型可以更准确地表征其自身的可靠性。这种证据感知策略为信任校准提供了更细粒度的基础，

response escalation, and offloading in collaborative LLM-SLM deployments, enabling more robust and semantically aligned inference.

Based on evidence-driven uncertainty estimation, future collaborative frameworks will shift from fixed rules to adaptive, intelligent policies. Reinforcement and meta-learning can optimize collaboration using task needs, resource limits, and real-time feedback. Cooperation will deepen from simple task splitting to joint optimization and shared representations, enabling finer coordination. In complex fields like multimodal and embodied intelligence, lightweight edge models will handle real-time sensing while cloud LLMs perform high-level reasoning, balancing latency and decision complexity. Challenges remain in ensuring robust collaboration, reducing communication costs, and enabling personalization. Future systems must balance performance, flexibility, and privacy for truly effective edge-cloud LLM-SLM collaboration.

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协作式 LLM-SLM 部署中的响应升级和卸载，从而实现更稳健且语义一致的推理。

基于证据驱动的不确定性估计，未来的协作框架将从固定规则转向自适应、智能政策。强化和元学习可以利用任务需求、资源限制和实时反馈来优化协作。合作将从简单的任务拆分深化到联合优化、共享表示，实现更精细的协调。在多模态和实体智能等复杂领域，轻量级边缘模型将处理实时传感，而云法学硕士则执行高级推理，平衡延迟和决策复杂性。在确保强有力的协作、降低通信成本和实现个性化方面仍然存在挑战。未来的系统必须平衡性能、灵活性和隐私，以实现真正有效的边缘云 LLM-SLM 协作。

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